

# IS THERE A PLATONIC TREE?

## CALIBRATING LATENT HYPERBOLICITY

## IN FOUNDATION MODELS

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### ABSTRACT

Hyperbolic methods for representation learning rest on a premise we call latent hyperbolicity: standard models are already tree-like, because their class geometry scores a low Gromov  $\delta$ . We show that this raw reading is confounded by dimension, by covariance spectrum and by the supremum statistic itself, and we build an instrument that reads every score as an excess over a random cloud with the same dimension and spectrum, ranks it against 200 matched replicates, and adds a depth test whose false-alarm rate and power are measured on real clouds. Applied to 12 vision backbones, 6 datasets and 16 text models, the instrument gives a nuanced answer. On image features, where the premise is read, the excess sits within null noise in most cells and reaches at most 40% of the null where it survives. On class centroids, clustered structure is genuine in 49 of 72 cells, but a star already produces it. Hierarchy above the superclasses is certified in 4 of 12 ImageNet backbones, the same four under a star with the real hubs' spectrum, by a test that never fires on real clouds with randomized hubs; a backbone trained in hyperbolic space lives in the near-flat regime and shows the same clustering and no detected depth. The trees are moderately shared: a naive comparison isolates the DINOv2 family as an island, an artifact of the clustering cut; once the cut is controlled a gap of about a third remains, every training recipe recovers the human taxonomy partially, more so with supervision, and the self-supervised tree lives in angles rather than distances. Read correctly, foundation models organize classes into clustered structure that is occasionally hierarchical and moderately shared; they do not converge to one common tree, and their raw tree-likeness is not evidence for hyperbolic geometry.

## 1 INTRODUCTION

figures/fig1\_concept.pdf (conceptual figure, drawn by the authors; same-size placeholder)

Figure 1: **A random cloud, a star and a tree all read as tree-like; only the instrument tells them apart.** Sketched: a high-dimensional random cloud, clusters around a hub and a nested hierarchy, all with a low raw reading, the excess that separates the cloud from the other two, and the depth test that separates the star from the tree.

Semantic knowledge is hierarchical, and hyperbolic space is the natural geometric home for hierarchies (Nickel & Kiela, 2017; Sarkar, 2011). A line of work therefore imposes it on representations

at training time, from vision to vision–language and language (Khurlov et al., 2020; Ermolov et al., 2022; Atigh et al., 2022; Bdeir et al., 2024; Desai et al., 2023; Pal et al., 2025; Chen et al., 2022; He et al., 2025; Sinha et al., 2024). These methods cite a premise, which we call *latent hyperbolicity*: standard models are said to organize concepts already in a tree-like metric structure, because the Gromov  $\delta$  of their features is low (Khurlov et al., 2020; Bdeir et al., 2024; Yang et al., 2025; Tifrea et al., 2019). Is the premise right, and what does it license?

The raw reading is confounded. Gromov  $\delta$  assigns zero to trees (Gromov, 1987), but three artifacts push it down without any hierarchy. High-dimensional clouds concentrate toward equidistance, which is a degenerate star tree: we call this the *dimension confound*. Anisotropic spectra mimic low-dimensional behavior: the *spectrum confound*. And the supremum over sampled quadruples is a one-quadruple extreme that does not converge: the *statistic confound*. The three are visible in real models: GPT-2 M sits at its matched null despite a low raw reading, and classic sentence embedders share their raw reading with vision-contrastive models while carrying none of their structure.

We build the *instrument* that reads  $\delta$  correctly. It calibrates the estimator on reference geometries, expresses every reading as an *excess* over a null cloud matched in dimension and covariance spectrum, ranks that excess against matched replicates, and adds a *depth test* whose false-alarm rate and power are measured on real clouds. With it we ask of geometry what convergence work asks of similarity (Huh et al., 2024; Gröger et al., 2026): what survives calibration, and is it shared? They calibrate similarity across models; we calibrate geometry within a model.

**The answer has three parts.** On image features the excess sits within null noise in most cells and is a fraction of the null where it survives, so the premise does not survive calibration as stated. On class centroids, clustered structure is genuine in 49 of 72 cells, but a star already produces it. Hierarchy above the superclasses is certified in 4 of 12 ImageNet backbones by a test that never fires on real clouds with randomized hubs, and a backbone trained in hyperbolic space shows the same clustering and no detected depth. The trees are moderately shared: the naive comparison manufactures an island, and once the clustering cut is controlled a moderate gap remains, every recipe recovers the human taxonomy partially, more so with supervision, and the self-supervised structure lives in angles.

#### Our contributions are:

1. **The instrument.** Estimator calibration, spectrum-matched nulls read as corrected ranks, and a depth test with measured false-alarm rate and power.
2. **The census.** Clustered structure in most models, hierarchy certified in a few, and a hyperbolic backbone as the control for imposing the geometry.
3. **The map of trees.** The island is an artifact of the cut; sharing is graded and supervision-dependent, and the self-supervised tree is angular.
4. **Consequences for practice.** What to measure before imposing curvature, and which zero-cost readout collects what is there.

## 2 RELATED WORK

Hyperbolic representation learning builds on tree embeddability and Poincaré embeddings and now spans vision, vision–language, graphs and language (Sarkar, 2011; Nickel & Kiela, 2017; Ganea et al., 2018; Chami et al., 2019). Its stated premise rests on a low raw reading for CNN and ViT features, token embeddings and word vectors, taken at the sample level and uncalibrated (Khurlov et al., 2020; Bdeir et al., 2024; Yang et al., 2025; Tifrea et al., 2019). Network science reads Gromov  $\delta$  on real graphs against reference graphs: the curvature of communication and social networks is compared with that of lattices, hyperbolic grids and Erdős–Rényi random graphs of comparable size, and  $\delta$  is read relative to the graph’s diameter (Narayan & Sanjeev, 2011; Kennedy et al., 2013). Adcock et al. (2013) pair  $\delta$  with tree decompositions and find the bare value of  $\delta$  too coarse a guide to tree-likeness on its own. Representation similarity, intrinsic dimension and neural collapse characterize Euclidean structure without testing tree-likeness (Kornblith et al., 2019; Ansuini et al., 2019; Pope et al., 2021; Pappas et al., 2020). The Platonic Representation Hypothesis argues that models converge toward shared representations; Gröger et al. (2026) show that permutation-calibrated similarity removes most of that convergence while local structure survives, and Park et al. (2024) read concept hierarchy in language models from orthogonality (Huh et al., 2024; Koepke et al., 2026).

What is new here is the instrument for embedding clouds rather than graphs: spectrum-matched nulls at the resolution of the null replicate, a depth test against matched stars measured on real clouds, a hyperbolic backbone as the control for imposing the geometry, and a calibration of the tree-to-tree comparison itself.

### 3 BACKGROUND AND PROBLEM SETUP

**Hypothesis under test (latent hyperbolicity).** Standard models organize concepts already in a tree-like metric structure, because the Gromov  $\delta$  of their features is low; on this reading a low raw  $\delta$  is evidence for hierarchy and licenses hyperbolic geometry.

#### 3.1 GROMOV $\delta$

**Definition 1** (four-point defect). For four points  $w, x, y, z$  of a metric space  $(X, d)$ , sort the three pairing sums  $d(w, x) + d(y, z)$ ,  $d(w, y) + d(x, z)$ ,  $d(w, z) + d(x, y)$  as  $S_{\max} \geq S_{\text{mid}} \geq S_{\min}$ . The four-point defect and the Gromov  $\delta$  of  $X$  are

$$\delta(w, x, y, z) = \frac{1}{2} (S_{\max} - S_{\text{mid}}), \quad \delta(X) = \sup_{w, x, y, z \in X} \delta(w, x, y, z). \quad (1)$$

Trees have  $\delta = 0$ , and under this definition the hyperbolic plane has  $\delta = \ln 2 \approx 0.693$ , distinct from the thin-triangles constant  $\ln(1 + \sqrt{2}) \approx 0.881$  (Nica & Špakula, 2016). We use the normalized  $\delta_{\text{norm}} = \delta_{\max} / \text{diam}$  throughout, on which trees < hyperbolic < spherical for hyperbolic regions of radius four and above (Table 2). Sampled-quadruple estimation has known bias and scale caveats, which the calibration-plus-null design addresses, and Ollivier–Ricci curvature on a kNN graph is the local companion (Fournier et al., 2015; Ollivier, 2009).

#### 3.2 ESTIMATION

**Definition 2** (reading of record). Let  $\hat{\delta}_{99.9}(X)$  be the 99.9th percentile of the defects of Eq. 1 over sampled quadruples of  $X$ , averaged over seeds. The reading of record is

$$\delta_{\text{norm}}(X) = \hat{\delta}_{99.9}(X) / \text{diam}(X). \quad (2)$$

The percentile is taken over half a million sampled quadruples per seed and averaged over ten seeds; the supremum, the pre-specified statistic, is reported in the appendix. The supremum did not converge under the quadruple budget on ImageNet and disagreed with the 99.9th percentile of the same defects in both directions. The reading is budget-stable under the record: across quadruple budgets the ImageNet excess of Section 4.2 drifts by less than half its own spread and never changes sign (Tables 3 and 4).

#### 3.3 OBJECTS AND NOTATION

The premise is read on the per-image features of frozen models, where the literature measures it. Hierarchy is measured on class centroids: in vision, the mean of each class’s cached training embeddings, a hundred per class on ImageNet; in text, the embeddings of “a photo of a {class}” over the ImageNet classes. A *cell* is one pair of model and class set; the census covers twelve vision backbones on six datasets and sixteen text models. The table below fixes the notation.

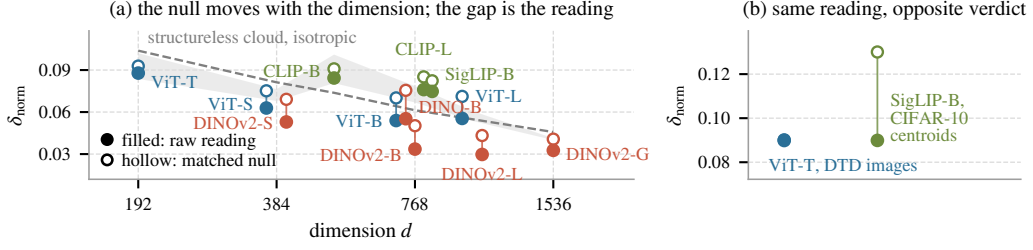


Figure 2: **The raw reading cannot be read on its own.** (a) Each ImageNet backbone’s raw reading (filled) and what a structureless cloud of the same dimension and spectrum reads (hollow); the nulls drift down with the dimension, and the segment between the two is the excess. (b) Two cells with the same raw reading and opposite verdicts: on ViT-T image features the null sits at the reading, on SigLIP-B centroids far above it.

| symbol                                      | meaning                                                                                                        |
|---------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| $X \in \mathbb{R}^{n \times d}$             | a cloud of $n$ points (images, or class centroids) in dimension $d$                                            |
| $C$                                         | number of classes of a cell; at the class level $n = C$                                                        |
| $K$                                         | number of null replicates per cell (Eq. 4)                                                                     |
| $\hat{\delta}_{99.9}, \delta_{\text{norm}}$ | the sampled 99.9th-percentile defect and the reading of record (Eq. 2)                                         |
| $e$                                         | the excess over the spectrum-matched null (Eq. 3); $e_B$ the same over the hub-randomizing null                |
| $r, p$                                      | percentile rank and left-tail $p$ of a cell (Eq. 4), corrected by Benjamini–Hochberg across the census (Eq. 5) |
| $z$                                         | the depth statistic in units of its spread (Eq. 6); a secondary summary elsewhere                              |

## 4 THE INSTRUMENT

If the premise were right, a low raw reading would be evidence on its own. The dimension confound says it is not, so this section builds the measurement layer in whose units every later claim is stated.

### 4.1 THE SPECTRUM-MATCHED NULL

Calibrated on reference geometries, the estimator recovers the tree zero and the  $\ln 2$  constant of  $\mathbb{H}^2$ , and spheres stay high. Iid Gaussians, which have no structure at all, drift lower as the dimension grows (Table 2). Figure 2a shows the same drift on the twelve ImageNet backbones: the null of each moves with its dimension, so the gap, not the level, is the reading. That drift is the dimension confound in its purest form: the raw reading of a real backbone must be compared with the reading of a structureless cloud of the same shape.

**Definition 3** (spectrum-matched null). Let  $X = U\Sigma V^\top$  be the singular value decomposition of the centered cloud. A null replicate is

$$X' = Q\Sigma V^\top, \quad Q \sim \text{Haar}(O(n)),$$

a cloud with the same  $n, d$ , mean and singular values as  $X$  and Haar-distributed, hence structureless, coefficients.

In plain terms, the spectrum null is a cloud with the same shape as the real one and no structure inside it; the hub null of Section 4.3 keeps every cluster of the real cloud and only rearranges where the clusters sit. The Gaussian-coefficient null does not reproduce the sample spectrum it claims to match when  $n$  is small relative to  $d$ , and it inflates the excess of low-rank clouds; the Haar null does reproduce it, so we use the Haar null and report both.

## 4.2 THE EXCESS AND ITS RANK

**Definition 4** (excess). With  $K$  null replicates  $X'_1, \dots, X'_K$  of Definition 3, the excess of a cell is

$$e(X) = \delta_{\text{norm}}(X) - \frac{1}{K} \sum_{k=1}^K \delta_{\text{norm}}(X'_k). \quad (3)$$

The primary evidence is not  $e$  but its rank among the replicates and the left-tail  $p$  that the rank gives,

$$r = \#\{k : \delta_{\text{norm}}(X'_k) > \delta_{\text{norm}}(X)\}, \quad p = \frac{1 + K - r}{K + 1}, \quad K = 200. \quad (4)$$

**Definition 5** (genuine). Order the  $m$  left-tail  $p$ -values of a census as  $p_{(1)} \leq \dots \leq p_{(m)}$ . A cell is genuine when its  $p$  is at or below the Benjamini–Hochberg threshold at level five per cent,

$$p \leq p_{(i^*)}, \quad i^* = \max\{i : p_{(i)} \leq 0.05 i/m\}, \quad (5)$$

and we use the word in this sense only.

A negative genuine excess certifies *clustered structure*; the word “tree-like” is reserved for the depth test of Section 4.3. A positive excess means less tree-like than chance given the spectrum (Appendix A.2). Figure 2b shows two cells with the same raw reading and opposite verdicts. Rank-based calibration is what gives permutation scores their finite-sample validity, so a  $z$ -score is kept only as a secondary summary, and the null is aggregation-aware in the sense of Gröger et al. (2026, §5): the same percentile of the same sampled defects is evaluated on every replicate.

## 4.3 THE DEPTH TEST

**Definition 6** (hub-randomizing null and excess B). Given a frame that assigns each centroid  $x_i$  to a cluster  $k(i)$  with hub  $h_{k(i)}$  (the mean of its cluster), write  $x_i = h_{k(i)} + o_i$ . The hub-randomizing null keeps every offset  $o_i$  and replaces the hub configuration by a Haar resample with the same spectrum,  $x'_i = h'_{k(i)} + o_i$ ; the excess of Eq. 3 computed against this null is  $e_B$ . Because the null keeps the clusters,  $e_B$  measures hierarchy above the labelled clusters only.

**Definition 7** (matched star and depth). The matched star of a cloud has Gaussian hubs at the real hub radius and, within each cluster, a Haar sample with the cluster’s own covariance, so that it carries the clustering of the real cloud and no hierarchy above it. With  $e_B$  read on the real cloud and on the star,

$$\text{depth} = e_B(X) - e_B(X_{\text{star}}), \quad z = \text{depth} / \text{s.d.}, \quad (6)$$

where the spread is taken over star seeds and null replicates, and the real cloud is certified as more hierarchical than a star when  $z \leq -2$ .

The frame is the WordNet cut on ImageNet and the coarse labels on CIFAR-100, and the star’s hubs are drawn either as Gaussians at the real hub radius or as a Haar resample of the real hubs. The test’s regime is measured rather than assumed: on synthetic hierarchies at ImageNet’s class count it meets a pre-set bar for power and false alarms, on the real ImageNet clouds it never fires when the hubs are randomized and misses a clean implanted tree at their noise level, and at CIFAR-100’s cluster sizes it is not validated (Section 6.3). It is therefore conservative and weak at ImageNet’s noise level: a positive verdict certifies and a negative one says nothing.

## 4.4 PROJECTION AND TASKS

A parameter-free map centers the features, scales the 95th-percentile norm to radius  $1/\sqrt{2}$  and preserves directions, so Euclidean, Poincaré and cosine readouts can be compared on the intact representation; nearest-centroid and 5-way 5-shot classification score them for Section 7 (Appendix A.12).

**Definitions at a glance.**

- *Four-point defect and  $\delta$* : the half-gap between the two largest pairing sums, and its supremum (Eq. 1).
- *Reading of record*: the 99.9th-percentile defect over the diameter (Eq. 2).
- *Spectrum-matched null*: the cloud with Haar coefficients and the real singular values (Definition 3).
- *Excess*: the reading minus the mean reading of the null replicates (Eq. 3).
- *Rank and  $p$* : replicates above the real value and the left-tail  $p$  they give (Eq. 4).
- *Genuine*: at or below the Benjamini–Hochberg threshold across the census (Eq. 5).
- *Depth and  $z$* : excess B of the real cloud minus that of its matched star, in units of the spread (Eq. 6).

## 5 EXPERIMENTAL SETUP

**Models and datasets.** Twelve vision backbones form the census: four supervised ViTs trained on ImageNet-21k, DINO and four DINOv2 sizes, and CLIP-B, CLIP-L and SigLIP-B; the ten with cached per-dataset features form the task grid. They are read on six class sets, ImageNet, CIFAR-100, CIFAR-10, DTD, FMNIST and MNIST, the first two with a real hierarchy. Sixteen text models read the thousand ImageNet class names: four GPT-2 sizes, three Pythia sizes, two OLMo sizes and seven sentence embedders. The controls are MERU and its Euclidean CLIP twins, two ViT-B trained on ImageNet-1k leaf labels, two self-supervised ResNet-50 backbones and the census ViT-B fine-tuned with and without a hierarchical objective; parameter counts are listed in Appendix A.14.

**Extraction.** Class centroids average a hundred cached training images per class on ImageNet and the full cached split on the other datasets. Vision backbones use the encoder’s default pooled embedding; causal language models use the last-token hidden state, extracted one prompt at a time, because left-padded batching alters GPT-2’s hidden states; text embedders use their recommended pooling, and the CLIP and SigLIP text towers their text encoders (Appendix A.10). The sample-level reading uses roughly a thousand stratified training images per cell for CIFAR-100 and DTD.

**Replicates, seeds and budgets.** Every cell is read against 200 Haar replicates, with half a million sampled quadruples per seed and ten seeds for the real cloud, and genuineness is decided by Benjamini–Hochberg over the vision census and over the text census separately. The depth test uses ten star seeds and the hub-randomizing null of Definition 6; its false-alarm rate and power are measured on synthetic hierarchies and on the real ImageNet clouds with implanted trees. The cosine reading repeats the census on L2-normalized centroids with geodesic distances and the same Haar null on the normalized cloud, as robustness only. With the instrument in place, we read the premise where it is read.

## 6 RESULTS

If standard models were latently hyperbolic, the instrument should find an excess on their image features, where the premise is read. It finds one within null noise in most cells, and what survives lives on class centroids, where a star already passes the census.

### 6.1 THE PREMISE AT THE SAMPLE LEVEL

**Finding 1.** The premise does not survive calibration: on image features the excess sits within null noise in most cells, and where it is genuine it is a fraction of the null reading, largest in the DINOv2 family.

We read the premise first on per-image features, roughly a thousand stratified training images per cell for CIFAR-100 and DTD, where the raw reading lies in the low band the literature reports. Calibrated, it is not genuine in most cells, and where it is genuine it removes between 7 and 39 per cent of the null reading, the larger figure for DINOv2-G on CIFAR-100 images, and it is largest in

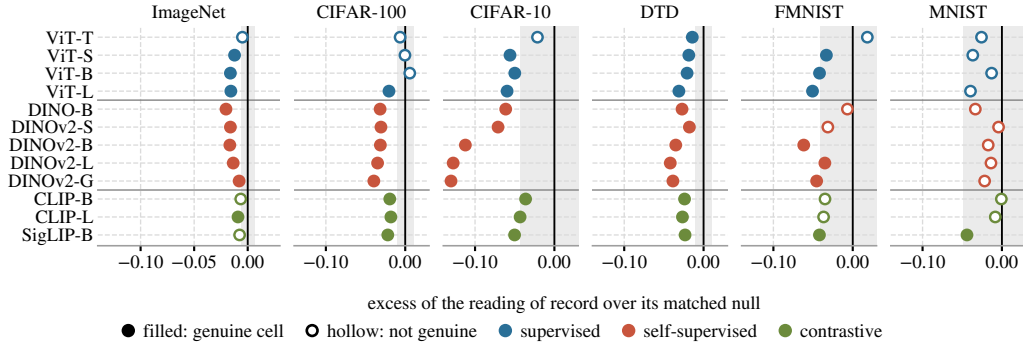


Figure 3: **Clustered structure is the rule at the class level.** Excess of the reading of record over its matched null, one panel per dataset, one marker per backbone; filled markers are genuine cells. Almost every marker sits left of zero, most are filled, the flat datasets scatter around zero, and the DINOv2 rows reach furthest left on CIFAR-10.

| model    | family | $\hat{\delta}_{99.9}$ IN | excess IN      | $r/200$ ( $p$ ) | exc. C100      | exc. C10       | $\rho_{WN}$ |
|----------|--------|--------------------------|----------------|-----------------|----------------|----------------|-------------|
| ViT-T    | sup.   | 0.088                    | $-0.005^\circ$ | 169 (.159)      | $-0.006^\circ$ | $-0.021^\circ$ | +0.52       |
| ViT-S    | sup.   | 0.063                    | $-0.012$       | 200 (.005)      | $-0.000^\circ$ | $-0.056$       | +0.50       |
| ViT-B    | sup.   | 0.054                    | $-0.016$       | 200 (.005)      | $+0.006^\circ$ | $-0.050$       | +0.49       |
| ViT-L    | sup.   | 0.055                    | $-0.016$       | 200 (.005)      | $-0.020$       | $-0.060$       | +0.53       |
| DINO-B   | SSL    | 0.055                    | $-0.020$       | 200 (.005)      | $-0.032$       | $-0.061$       | +0.36       |
| DINOv2-S | SSL    | 0.053                    | $-0.016$       | 200 (.005)      | $-0.031$       | $-0.071$       | +0.22       |
| DINOv2-B | SSL    | 0.034                    | $-0.017$       | 200 (.005)      | $-0.031$       | $-0.112$       | +0.18       |
| DINOv2-L | SSL    | 0.030                    | $-0.014$       | 200 (.005)      | $-0.035$       | $-0.128$       | +0.18       |
| DINOv2-G | SSL    | 0.033                    | $-0.008$       | 200 (.005)      | $-0.040$       | $-0.130$       | +0.20       |
| CLIP-B   | contr. | 0.084                    | $-0.007^\circ$ | 172 (.144)      | $-0.019$       | $-0.036$       | +0.57       |
| CLIP-L   | contr. | 0.076                    | $-0.009$       | 197 (.020)      | $-0.018$       | $-0.043$       | +0.59       |
| SigLIP-B | contr. | 0.075                    | $-0.008^\circ$ | 189 (.060)      | $-0.022$       | $-0.050$       | +0.58       |

Table 1: **Clustered structure is the rule at the class level.** One row per backbone.  $\hat{\delta}_{99.9}$ : raw ImageNet reading; excess:  $\hat{\delta}_{99.9}$  minus the mean of 200 Haar spectrum-matched null replicates;  $r/200$ : replicates above the real value, with its left-tail  $p$ ; genuine: Benjamini-Hochberg-corrected  $p \leq 0.05$  over the 72 cells;  $^\circ$ : not genuine; bold: less clustered than the null.  $\rho_{WN}$ : Spearman correlation of inter-centroid and WordNet distances. The same reading on per-image features is genuine in 10 of 24 cells (Table 5). Under image resampling and estimator noise the genuine count is 42–47 of 72 (Table 4). Other datasets and the four null  $\times$  statistic verdicts: Table 3; task columns: Table 13.

the DINOv2 family, whose structure is the angular tree of Section 6.4, which cosine collects. This is the reading that motivates hyperbolic methods, and calibrated it is what a random cloud of the same dimension and spectrum produces, up to a fraction of the null (Table 5).

## 6.2 CLUSTERED STRUCTURE AT THE CLASS LEVEL

**Finding 2.** On class centroids, clustered structure is genuine in most cells of the census and in every family; the count survives resampling, and a star already produces it.

**Clustered structure is the rule.** Table 1 gives the census of record. Figure 3 shows it: almost every marker sits left of zero, most are filled, the flat datasets scatter around it, and the DINOv2 rows reach furthest left on CIFAR-10. In 49 of the 72 cells the gap is one that chance does not produce. On ImageNet and CIFAR-100, the two sets with a real hierarchy, the count is 18 of 24. The structure belongs to no single family or architecture: every family contributes genuine cells and

two self-supervised ResNets read the same as the ViTs. What moves the reading is the label set: the smallest backbone is genuine on one dataset only, the flat datasets sit at the resolution limit of ten classes, and a class-count control shows that the reading follows the hierarchy of the labels rather than their number (Table 4).

**The count survives resampling.** The census null accounts for replicate variability but not for image resampling or the estimator’s seeds. Folding all three into one spread, or repeating the census on thirty resampled centroid sets, leaves most of the count in place, and the cells that move are on the transfer sets (Table 1).

### 6.3 DEPTH ABOVE THE LABELLED CLUSTERS

**Finding 3.** A star passes the census, so a negative excess says clustered, not deep. Hierarchy above the superclasses is certified in four ImageNet backbones under both matched stars by a test that never fires on real clouds with randomized hubs, and imposing the geometry does not create detected depth.

**A star passes the census.** The excess measures structure beyond second moments, and a pure star already has such structure. A synthetic star of thirty clusters with no hierarchy scores an excess as deep as any real cell. We call this the *star caveat*. The excess therefore measures how classes are arranged, not the geometry they are embedded in, and a negative excess says clustered, not deep. Whether the arrangement is deeper than a star is a separate question, which needs a test of its own (Table 2).

**Depth is certified on ImageNet and not on CIFAR-100.** On ImageNet the real cloud is more hierarchical than its matched star in 4 of 12 backbones, three supervised ViTs and DINOv2-L, and never less. The other eight are not detected, which is not the same as not hierarchical. For the four certified backbones the same clusters with random hubs raise no alarm in any zero-strength run, so the verdict comes from their real hub arrangement, which the test reads as deeper than the implanted two-level tree. The certified set shifts with the choice of frame, as Table 6 shows. CIFAR-100, with clusters of a few points, lies outside the validated regime. An earlier isotropic star produced false alarms and is withdrawn (Figure 4a).

**A spectrum-matched star confirms the four.** We repeated the test with the star’s hubs drawn as a Haar resample of the real hubs, so that the star carries the exact hub spectrum. The certified set is the same four, with  $z$  from  $-2.0$  to  $-3.7$ . On the real clouds with implanted trees the new star raises 0 of 60 false alarms at zero strength and detects one implant in sixty at full strength. Read instead as a rank over the ten star seeds, at a resolution of one part in eleven, the verdict places the real cloud below every star seed for the certified four and for most of the others alike, so the rank does not separate certified from uncertified backbones while  $z$  does. Supervision on leaf labels alone can produce this depth and need not: of two ViT-B trained on ImageNet-1k leaf labels, the augreg recipe is certified under both stars and DeiT-B is not, so the IN-21k label hierarchy is not what makes the census ViTs deep (Table 6).

**On real clouds the test is conservative and weak.** We measured the test on the real ImageNet clouds themselves, replacing the hub arrangement by an implanted two-level tree of graded strength while keeping every within-cluster offset, and the census cannot tell the implanted tree from the star. The test raises no false alarm, declaring none of the sixty zero-strength runs hierarchical. A clean implanted tree is detected in none of the twelve backbones, and the misses are a matter of noise level: the within-cluster spread of the real clouds is between one and four times their between-hub spread, the synthetic sweep covered ratios below one, and once the real offsets are shrunk into that range every backbone is detected, and two contrastive backbones also fire at zero strength, so the shrunk variant is a diagnostic of power, not a substitute test. The test is therefore conservative and weak at ImageNet’s noise level: its positive verdicts stand, and a backbone it does not certify is not thereby flat. A trained control, a ViT-B fine-tuned with a hierarchical objective, is inconclusive (Table 7, Figure 4b).

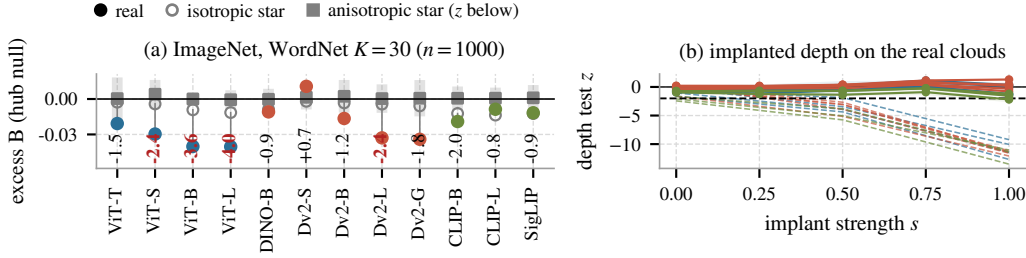


Figure 4: **The depth test certifies hierarchy above the superclasses in 4 of 12 ImageNet backbones, raises no false alarm on real clouds, and misses implanted depth at ImageNet’s noise level.** Plotted: (a) the excess under the hub-randomizing null for the real ImageNet centroids and for their matched stars, hollow for isotropic and square for anisotropic, with the  $z$  of real minus anisotropic star under each backbone, red at  $z \leq -2$ ; (b) the  $z$  of the same test after replacing the hub arrangement by an implanted two-level tree of strength  $s$ , dashed with the within-cluster spread shrunk into the validated range.

**Imposing the geometry does not create detected depth.** MERU embeds images and text on the Lorentz hyperboloid with an entailment objective, and its released Euclidean twin is a CLIP baseline with the same backbone, data and recipe minus the hyperbolic lift, so the pair isolates the geometry (Desai et al., 2023). Its embeddings sit in the near-flat regime of the hyperboloid, at a Lorentz-to-Euclidean distance ratio of 0.997, so the control tests the training objective rather than a curved geometry. Passed through the instrument unchanged, the two read the same excess on ImageNet images, and MERU’s reading in its own Lorentz metric equals its Euclidean reading to the fourth decimal. Under the depth test, conservative and weak at this noise level, neither model is detected as more hierarchical than a matched star, and MERU is never ahead of its twin. Training in hyperbolic space therefore neither creates the clustering nor adds detected depth (Table 6).

**Neural collapse is the flat limit, not the reading.** Neural collapse predicts that the class means of a supervised classifier converge to a simplex equiangular tight frame (Pappan et al., 2020). An exact frame is the extreme case of the star caveat, and our reading differs from it in two ways. The arrangement we certify is not flat, because class means cluster around superclass hubs that a frame would erase, and cutting the centroid dendrograms recovers the CIFAR-100 superclasses. The same clustered structure also appears without class labels, on transfer datasets and in causal language models, outside the setting where neural collapse is stated, and the depth test asks what neural collapse does not, whether there is hierarchy above the hubs (Table 8).

**Scale deepens the excess for one family only.** Scale does not deepen the excess in general, with one exception: the DINOv2 rows of Figure 3 reach further left with size on CIFAR-10 and, weakly, on CIFAR-100. Training moves the raw reading within an architecture: fine-tuning on a task with no class hierarchy raises it, it falls across transformer depth, and shuffling the embedding-to-label correspondence leaves it unchanged (Table 8). What survives at the class level is clustered structure, genuine in most cells, rarely deep, and not created by the embedding geometry. Whether that structure is shared across models, and whose tree it is, is the question we turn to next.

#### 6.4 WHOSE TREE

**Finding 4.** The trees are moderately shared: the categorical island is an artifact of the clustering cut and a moderate gap remains, alignment with the human taxonomy follows supervision and recipe rather than tree-likeness, and the self-supervised tree lives in angles.

If the measured trees were a shared representation of reality, models should agree with each other and with human taxonomy. Whether they appear to depends on how the trees are measured.

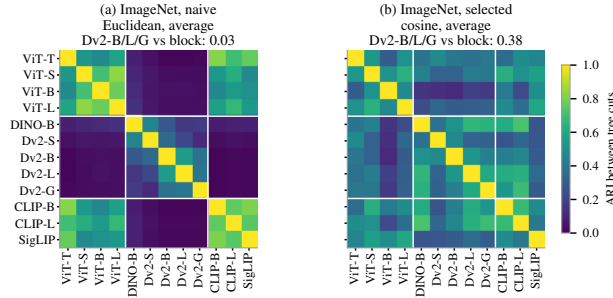


Figure 5: **The categorical island is a chaining artifact; about a third of the within-block agreement is missing under most configurations.** Plotted: pairwise ARI between the dendrogram cuts of the 12 vision models on ImageNet under the naive configuration (a) and the selected one (b), with the mean DINOv2-vs-block ARI above each panel.

**The naive map manufactures an island.** We extract each model’s hierarchy as a dendrogram over the shared centroids and compare hierarchies by the adjusted Rand index between cut partitions and by the cophenetic correlation between tree and distances. Under the naive configuration, Euclidean distances with average linkage, the map is spectacular: supervised ViTs and contrastive VLMs form one block, and the label-free DINOv2 family appears as an island that agrees with no supervised model, which we call the *island*. Its cuts are degenerate: DINOv2’s Euclidean dendrograms chain, almost every class falls into one cluster, and the index is forced to zero by construction. A configuration must therefore be selected by a criterion fixed without reference to the agreements being compared: we require the highest cophenetic fidelity among configurations whose reference cuts are non-degenerate for every model, which selects cosine-average on ImageNet and cosine-complete on CIFAR-100. The criterion is applied per dataset, because degeneracy is a property of the dataset, and it never sees the agreements it selects for. Under every non-degenerate configuration the gap stays moderate and never returns to the island. Figure 5 shows the map before and after this control (Table 9).

**The categorical island is an artifact, and a moderate gap remains.** Under the selected configurations three measures agree. At the cut, the island shrinks from categorical to moderate: the DINOv2 family agrees with the block at 0.38 against 0.48 within the block, and under the other admissible configurations about a third of the within-block agreement is missing. In cophenetic correlation a gap remains under every configuration, and in triplet agreement, which needs no cut, it closes under cosine-average and persists under the other configurations, the Euclidean ones included. Across every non-degenerate configuration, label-free models recover the human taxonomy partially and supervision deepens the resemblance on average across the supervised backbones tested: paired within each admissible configuration, the label-supervised ViT recovers more than each DINOv2 model in all but one of twelve comparisons (Table 9).

**The self-supervised tree lives in angles.** DINOv2’s semantics concentrate in the angular component of its representation: for DINOv2-L, sibling triplets are resolved far better under cosine than under Euclidean distance. We call this structure the *angular tree*. It is why DINOv2’s Euclidean dendrograms chain into the island and why the triplet gap closes only under cosine-average. The census agrees when we repeat it on cosine geometry, with L2-normalized centroids, geodesic distances and the same Haar null on the normalized cloud: the verdict matches the Euclidean census in 65 of 72 cells, and DINOv2 on ImageNet is genuine under both readings. We report the cosine census as robustness, never as the record (Table 3).

**WordNet alignment follows supervision and recipe, not tree-likeness.** The correlation between inter-centroid distances and WordNet tree distances on ImageNet orders the families by supervision: highest for the contrastive VLMs, then the supervised ViTs, and lowest for DINOv2, with shuffle and Gaussian controls at zero. Leaf-label supervision does not guarantee the alignment: the augreg ViT-B trained on ImageNet-1k is aligned like the IN-21k ViTs, while DeiT-B is nearly unaligned yet recovers the CIFAR-100 superclasses as well as any backbone, so alignment is recipe-dependent

even among leaf-supervised ViTs. DINOv2 is nevertheless genuine on ImageNet and both CIFAR sets and recovers the true CIFAR-100 superclasses from its dendrograms, against zero for permuted labels. Across the three families, WordNet content and robustness of the excess anti-correlate, as Table 1 shows: the supervised and contrastive families are the most aligned and the most fragile, DINOv2 the least aligned and the most robust. Alignment with the human taxonomy is therefore something training puts in, not something tree-likeness reveals. On DBpedia Classes, a three-level ontology independent of WordNet, no configuration degenerates, and the three text embedders recover the true level-2 partition in every configuration and agree with each other. Their excess there is genuine with every replicate above the real value, so the recovered hierarchy is real, shared and configuration-robust (Table 10).

**The recovery is not WordNet circularity.** ImageNet classes are WordNet synsets, so recovering WordNet there could be circular. Three controls bound the circularity: WordNet-coherent groupings read a lower raw  $\delta$  than random ones in all but one of twelve models, single-image distances still correlate with WordNet, and the decoupling of alignment from robustness is the opposite of what label injection would produce (Table 10).

**In text, clustered structure depends on recipe, scale and probe.** On the thousand ImageNet class names the sentence embedders are not genuine, despite raw readings indistinguishable from vision backbones whose excess is. Within causal language models on the same probe, GPT-2 L and XL are genuine under every construction of the null and statistic and under cosine, GPT-2 S is genuine at the margin under the record and flips under the supremum and under cosine, and GPT-2 M is not genuine under the record: robust at the larger scales, fragile at the smaller. Pythia and OLMo-1B are genuine at every scale tested, and extraction protocol and prompt template move the fragile values but not the scale ordering (Table 11).

## 6.5 LOCAL AGREEMENT ACROSS MODELS

We apply the permutation calibration of Gröger et al. (2026) to the sixty-six model pairs, and mutual-kNN agreement and centroid CKA both survive with every pair significant, while the Poincaré reading improves neither (Table 12). What a practitioner can collect from this local agreement is the subject of the next section. Centroids distill a relational base that models share locally, in who is near whom, even where their trees differ.

## 7 IMPLICATIONS FOR HYPERBOLIC REPRESENTATION LEARNING

If the raw reading were evidence, a practitioner could set the curvature from it. It is not, so this section states what to read before imposing curvature and which readout collects the structure that survives.

**The raw reading cannot select a curvature.** Khrulkov et al. (2020) set the curvature of the Poincaré ball from the measured relative hyperbolicity,

$$c = (0.144/\delta_{\text{rel}})^2, \quad \delta_{\text{rel}} = 2\delta/\text{diam}. \quad (7)$$

Since the raw reading is dominated by the dimension and spectrum confounds, such a curvature tracks the backbone’s width rather than its hierarchy. Evaluated on the Gaussian band of Table 2, Eq. 7 would give structureless clouds a curvature from 0.48 at  $d=192$  to 2.5 at  $d=1536$ . The excess is the quantity to read instead: on ImageNet centroids it removes 5 to 33 per cent of the null reading, many times the null’s own spread, and on image features it sits within noise in most cells (Table 1).

**The calibrated reading predicts the zero-cost gain.** Within each hierarchical dataset, the record excess predicts the nearest-centroid gain of the Poincaré readout on all four datasets and the few-shot gain on three, with confidence intervals that exclude zero and a family control that leaves the within-dataset relation in place. The depth verdict predicts no gain anywhere. The prediction is a diagnostic read off the families it is scored on, not a policy, and cosine remains the safe default (Table 13, Figure 11).

**Measure before imposing.** For a frozen backbone the instrument answers three questions in minutes. Is there structure beyond a spectrum-matched null, and is it deeper than a star? The excess and its rank answer the first, and the depth test the second, conservative and weak at ImageNet’s noise level, so that a positive verdict certifies and a negative one says nothing. Where the structure lives, in angles or in distances, the cosine reading of Section 6.4 says.

**A zero-cost readout collects what is there.** Changing the readout metric costs nothing, and cosine collects the angular tree. The Poincaré readout adds from +0.9 to +1.3 pp over cosine for the contrastive VLMs on the transfer sets, while for the other families its effect is inconsistent in sign. On flat-label datasets no metric helps, and sample-level tasks never benefit (Table 13).

## 8 DISCUSSION AND LIMITATIONS

**What the paper establishes.** Read correctly, foundation models organize classes into clustered structure that is occasionally hierarchical and moderately shared; they do not converge to one common tree, and their raw tree-likeness is not evidence for hyperbolic geometry. In three acts: the premise of latent hyperbolicity is confounded and its sample-level reading sits within null noise; what survives at the class level is clustered structure in 49 of 72 cells, hierarchy above the superclasses in 4 of 12 ImageNet backbones, no detected depth from imposing the geometry, and a moderately shared, supervision-dependent recovery of the human taxonomy, angular for the self-supervised family; and what follows is that curvature should be measured rather than assumed.

**Open questions.** Why the structure is clustered but rarely certified as deep, and why the self-supervised tree is angular, are open: every recipe recovers the superclass hubs, yet only four backbones place them detectably beyond a star.

**Limitations.** (i) The excess certifies structure beyond second moments, star-like clustering included; the depth test certifies hierarchy above its frame only, is conservative and weak at ImageNet’s noise level, leaves hierarchy within the clusters untested, and its one trained positive control is inconclusive (Table 7). (ii) The census covers class-centroid geometry only. (iii) Causal-LM values, not their scale ordering, are template-conditioned. (iv) The objective–geometry link is correlational: it rests chiefly on DINOv2’s scale range, and two leaf-label ViTs show that supervision can produce depth and alignment without guaranteeing either, as Table 6 reports. (v) Downstream scope is prototype-based metric tasks with modest gains. (vi) The descriptor  $\xi$  tracks norm dispersion rather than curvature, as Table 14 reports. (vii) A causal LM’s class representation depends on the extraction protocol as much as on the model. (viii) The sample-level reading covers two datasets and one image budget.

### ETHICS STATEMENT

This work analyzes publicly released models and standard public datasets, involves no human subjects and no personal data, and proposes an analysis methodology and a metric-selection heuristic. We do not foresee ethical risks beyond those already associated with the underlying models and datasets.

### REPRODUCIBILITY STATEMENT

Every number in the paper is annotated in the source with the result file that produces it; all tables are generated by script from those files (Table 16), and the full pipeline (feature extraction, diagnostics, nulls, calibration, tasks, statistics, figures) with fixed seeds will be released. Estimator settings are stated in §4 and the model panel in Table 15. The instrument ships as one script in the anonymized repository (`TODO(author)`) that reproduces any cell of Table 1 from its centroid matrix; its output on two cells is checked against the table at every freeze. Computing  $\delta_{\text{norm}}$  on 1000 centroids takes seconds on CPU; the full task grid runs in under 30 minutes on one consumer GPU.

## AI USE STATEMENT

In this work, we used generative AI tools for writing assistance (drafting and editing text under continuous author direction), for code generation (analysis scripts and the experiment pipeline, run on the authors' hardware) and for data analysis assistance. We have not used generative AI tools for research ideation or for producing experimental results, and the remaining tasks with required disclosure are not applicable to this work. We have reviewed all AI-assisted work: every number in the paper is regenerated by script from the released result files and checked by an automated sweep at every freeze, LLM-generated code was verified and tested for correctness by the authors, and all experimental designs, results, verifications and final claims were checked and approved by the authors. We take responsibility for the final content of this work, including text, claims or artifacts produced with the aid of generative AI.

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## A APPENDIX: ONE TABLE PER QUESTION

Every table is generated by script from the released per-cell result files, and no value is hand-transcribed; the tables are numbered in the order the main text first cites them, one per question, with their result files indexed in Table 16.

### A.1 IS A LOW RAW READING EVIDENCE? CALIBRATION ON REFERENCE GEOMETRIES AND SYNTHETIC CLUSTERS

(a) Reference geometries.

| space                                                   | absolute $\delta$   | $\delta_{\text{norm}}$              |
|---------------------------------------------------------|---------------------|-------------------------------------|
| balanced binary tree (depth 10)                         | 0.000               | 0.000                               |
| $\mathbb{H}^2$ ( $K = -1$ ), region radius $R=2/4/8/16$ | 0.65/0.69/0.69/0.69 | 0.162/0.087/0.043/0.022             |
| uniform $S^{99}$ (chord / geodesic)                     | 0.25/0.37           | 0.143/0.179                         |
| iid Gaussian ( $n=1000$ ), $d=192, \dots, 1536$         | —                   | 0.104/0.081/0.074/0.061/0.055/0.046 |

(b) Synthetic clusters: what each null certifies.

| synthetic cloud ( $n=1000, d=768$ )  | $\hat{\delta}$ | spectrum null A |        | hub-randomizing null B |        |
|--------------------------------------|----------------|-----------------|--------|------------------------|--------|
|                                      |                | Gaussian        | Haar   | Gaussian               | Haar   |
| 30-cluster star, tight (0.1)         | 0.055          | −0.115          | −0.115 | −0.126                 | −0.058 |
| 30-cluster star, mid (0.3)           | 0.059          | −0.104          | −0.103 | −0.114                 | −0.049 |
| 30-cluster star, loose (0.6)         | 0.061          | −0.084          | −0.082 | −0.090                 | −0.033 |
| 2-level hierarchy $6 \times 5$ (0.3) | 0.044          | −0.143          | −0.143 | −0.113                 | −0.112 |

Table 2: **A low raw  $\delta$  is not evidence of hierarchy, and the spectrum excess certifies clustering, not depth.** (a) Calibration on reference geometries: the estimator recovers the tree zero and the four-point constant of  $\mathbb{H}^2$ , spheres stay high, and iid Gaussians drift low as  $d$  grows, the dimension confound; the ordering trees  $<$  hyperbolic  $<$  spherical holds for hyperbolic regions of radius  $R \geq 4$ . Gaussian row from the released control file; tree/ $\mathbb{H}^2$ /sphere rows re-verified synthetically at every freeze. (b) Excess of synthetic clouds (means over seeds; within/between noise ratio in parentheses) under the spectrum null A, with Gaussian coefficients and with Haar-rotated coefficients (exact sample spectrum), and under the hub-randomizing null B, which keeps every cluster and resamples the hub configuration. A pure star sits far below null A under either construction, so null A certifies clustering, not depth; null B also reports the star as hierarchical, since a near-regular simplex of hubs already minimizes  $\delta$ , which is why the depth test of Table 6 is read against a matched star rather than against null B. Under the Haar construction the two-level hierarchy sits 2–3 $\times$  below the star; under the plain Gaussian null the mid-radius star scores −0.104 and the  $6 \times 5$  hierarchy −0.14, a fraction of either’s excess.

### A.2 IS THE CLUSTERED STRUCTURE GENUINE? THE VISION CENSUS UNDER EVERY CONSTRUCTION

**What each null preserves.** The spectrum-matched null, Gaussian or Haar, preserves  $n$ ,  $d$ , the mean and the covariance spectrum and destroys all higher-order structure; the PC-permutation null preserves each principal component’s marginal and destroys the dependence across components; the label shuffle keeps the point cloud itself and destroys the embedding–label correspondence; and the hub-randomizing null keeps each cluster’s geometry and hub assignment and destroys the hub arrangement, which is why it serves the depth test only when read against a matched star. “Genuine” in the text always means beyond the first of these. In plain terms, the spectrum null is a cloud with the same shape as the real one and no structure inside it; the hub null of Section 6 keeps every cluster of the real cloud and only rearranges where the clusters sit.

(a) The census of record per cell: Haar spectrum-matched null, 99.9th-percentile statistic, 200 replicates.

| model    | IN      |     |      | C100    |     |      | C10     |     |      | DTD    |     |      | FMNIST  |     |      | MNIST   |     |      |
|----------|---------|-----|------|---------|-----|------|---------|-----|------|--------|-----|------|---------|-----|------|---------|-----|------|
|          | exc.    | r   | p    | exc.    | r   | p    | exc.    | r   | p    | exc.   | r   | p    | exc.    | r   | p    | exc.    | r   | p    |
| ViT-T    | -0.005° | 169 | .159 | -0.006° | 151 | .249 | -0.021° | 172 | .144 | -0.014 | 197 | .020 | +0.018° | 34  | .831 | -0.026° | 168 | .164 |
| ViT-S    | -0.012  | 200 | .005 | -0.000° | 111 | .448 | -0.056  | 200 | .005 | -0.019 | 199 | .010 | -0.033  | 196 | .025 | -0.037° | 181 | .100 |
| ViT-B    | -0.016  | 200 | .005 | +0.006° | 43  | .786 | -0.050  | 200 | .005 | -0.021 | 200 | .005 | -0.042  | 200 | .005 | -0.013° | 142 | .294 |
| ViT-L    | -0.016  | 200 | .005 | -0.020  | 200 | .005 | -0.060  | 200 | .005 | -0.031 | 200 | .005 | -0.051  | 200 | .005 | -0.040° | 193 | .040 |
| DINO-B   | -0.020  | 200 | .005 | -0.032  | 200 | .005 | -0.061  | 200 | .005 | -0.027 | 200 | .005 | -0.007° | 127 | .368 | -0.034° | 188 | .065 |
| DINOv2-S | -0.016  | 200 | .005 | -0.031  | 200 | .005 | -0.071  | 200 | .005 | -0.018 | 200 | .005 | -0.031° | 193 | .040 | -0.004° | 112 | .443 |
| DINOv2-B | -0.017  | 200 | .005 | -0.031  | 200 | .005 | -0.112  | 200 | .005 | -0.035 | 200 | .005 | -0.062  | 200 | .005 | -0.017° | 147 | .269 |
| DINOv2-L | -0.014  | 200 | .005 | -0.035  | 200 | .005 | -0.128  | 200 | .005 | -0.042 | 200 | .005 | -0.035  | 197 | .020 | -0.014° | 132 | .343 |
| DINOv2-G | -0.008  | 200 | .005 | -0.040  | 200 | .005 | -0.130  | 200 | .005 | -0.039 | 200 | .005 | -0.046  | 199 | .010 | -0.022° | 158 | .214 |
| CLIP-B   | -0.007° | 172 | .144 | -0.019  | 200 | .005 | -0.036  | 195 | .030 | -0.024 | 200 | .005 | -0.035° | 181 | .100 | -0.001° | 104 | .483 |
| CLIP-L   | -0.009  | 197 | .020 | -0.018  | 200 | .005 | -0.043  | 199 | .010 | -0.027 | 200 | .005 | -0.037° | 188 | .065 | -0.008° | 126 | .373 |
| SigLIP-B | -0.008° | 189 | .060 | -0.022  | 200 | .005 | -0.050  | 199 | .010 | -0.023 | 200 | .005 | -0.042  | 198 | .015 | -0.044  | 198 | .015 |

(b) Verdict under the four null  $\times$  statistic constructions and the excess as a fraction of the null reading.

| model    | IN       | C100     | C10      | DTD      | FMNIST   | MNIST    | sup. IN exc. (r) |
|----------|----------|----------|----------|----------|----------|----------|------------------|
| ViT-T    | ●●● -5%  | ●●● -5%  | ○○○ -15% | ●●● -11% | ○○○ +14% | ○○○ -16% | -0.020 (199)     |
| ViT-S    | ●●● -16% | ●●● -0%  | ●●● -39% | ●●● -16% | ●●● -25% | ○○○ -22% | -0.020 (200)     |
| ViT-B    | ●●● -23% | ○○○ +5%  | ●●● -36% | ●●● -19% | ●●● -30% | ○○○ -9%  | -0.022 (200)     |
| ViT-L    | ●●● -22% | ●●● -18% | ●●● -42% | ●●● -29% | ●●● -35% | ○○○ -24% | -0.027 (200)     |
| DINO-B   | ●●● -27% | ●●● -28% | ●●● -41% | ●●● -22% | ○○○ -5%  | ○○○ -24% | +0.000 (96)      |
| DINOv2-S | ●●● -23% | ●●● -31% | ●●● -46% | ●●● -15% | ○○○ -23% | ○○○ -3%  | +0.017 (0)       |
| DINOv2-B | ●●● -33% | ●●● -39% | ●●● -71% | ●●● -31% | ●●● -43% | ○○○ -11% | +0.018 (0)       |
| DINOv2-L | ●●● -31% | ●●● -49% | ●●● -82% | ●●● -38% | ●●● -25% | ○○○ -8%  | +0.006 (0)       |
| DINOv2-G | ●●● -20% | ●●● -57% | ●●● -84% | ●●● -35% | ●●● -31% | ○○○ -13% | +0.020 (0)       |
| CLIP-B   | ●●● -7%  | ●●● -16% | ●●● -26% | ●●● -20% | ○○○ -20% | ○○○ -1%  | -0.023 (199)     |
| CLIP-L   | ●●● -11% | ●●● -16% | ●●● -30% | ●●● -21% | ○○○ -24% | ○○○ -5%  | -0.016 (198)     |
| SigLIP-B | ○○● -9%  | ●●● -18% | ●●● -36% | ●●● -20% | ●●● -27% | ●●● -28% | -0.015 (194)     |

Table 3: **Clustered structure is the rule at the class level under every construction of the null and statistic, and the verdict does not depend on the metric.** (a) Per cell: excess of  $\hat{\delta}_{99.9}$  over the null mean,  $r$  = replicates above the real value,  $p = (1 + \#\{\text{null} \leq \text{real}\})/201$ ; genuine = Benjamini-Hochberg-corrected  $p \leq 0.05$  over the 72 cells (°: not genuine). Genuine cells: 49/72 overall, 18/24 on ImageNet+CIFAR-100; sign agreement with the original 5-replicate Gaussian  $\times$  supremum census on the 72 comparable cells: 67/72. (b) Four symbols per cell in the order Haar  $\times$  p99.9 (the record), Haar  $\times$  supremum, Gaussian  $\times$  p99.9, Gaussian  $\times$  supremum, ● = genuine under that construction; genuine counts 49/72, 46/72, 52/72, 52/72. The percentage is the record excess divided by the mean null reading (ImageNet -33% to -5%; all cells -84% to +14%); the last column is the ImageNet excess of the supremum statistic under the same Haar null with its rank. The Haar construction reproduces the sample spectrum exactly; the Gaussian one does not at small  $n$  and inflates excesses for low-rank clouds (Table 4). The supremum is decided by a few extreme quadruples and disagrees with the 99.9th percentile in both directions on DINOv2-S/B/G ImageNet.

(c) The same census on cosine geometry (L2-normalized centroids, geodesic distances, Haar null on the normalized cloud): robustness, not the record.

| model    | IN     |     | C100                |     | C10    |     | DTD    |     | FMNIST              |     | MNIST               |     |
|----------|--------|-----|---------------------|-----|--------|-----|--------|-----|---------------------|-----|---------------------|-----|
|          | exc.   | $r$ | exc.                | $r$ | exc.   | $r$ | exc.   | $r$ | exc.                | $r$ | exc.                | $r$ |
| ViT-T    | -0.012 | 200 | -0.014 <sup>o</sup> | 191 | -0.043 | 198 | -0.017 | 199 | +0.007 <sup>o</sup> | 73  | -0.026 <sup>o</sup> | 167 |
| ViT-S    | -0.017 | 200 | -0.017              | 200 | -0.073 | 200 | -0.021 | 200 | -0.049              | 200 | -0.036 <sup>o</sup> | 183 |
| ViT-B    | -0.022 | 200 | -0.006 <sup>o</sup> | 170 | -0.066 | 200 | -0.021 | 200 | -0.041              | 199 | -0.010 <sup>o</sup> | 130 |
| ViT-L    | -0.021 | 200 | -0.024              | 200 | -0.080 | 200 | -0.024 | 200 | -0.066              | 200 | -0.043              | 196 |
| DINO-B   | -0.021 | 200 | -0.035              | 200 | -0.070 | 200 | -0.024 | 200 | -0.006 <sup>o</sup> | 116 | -0.038 <sup>o</sup> | 193 |
| DINOv2-S | -0.014 | 200 | -0.033              | 200 | -0.084 | 200 | -0.018 | 200 | -0.054              | 200 | -0.004 <sup>o</sup> | 113 |
| DINOv2-B | -0.013 | 200 | -0.032              | 200 | -0.140 | 200 | -0.021 | 200 | -0.081              | 200 | -0.019 <sup>o</sup> | 147 |
| DINOv2-L | -0.010 | 200 | -0.029              | 200 | -0.151 | 200 | -0.024 | 200 | -0.051              | 200 | -0.020 <sup>o</sup> | 145 |
| DINOv2-G | -0.006 | 200 | -0.029              | 200 | -0.169 | 200 | -0.017 | 200 | -0.064              | 200 | -0.023 <sup>o</sup> | 157 |
| CLIP-B   | -0.017 | 200 | -0.023              | 200 | -0.039 | 197 | -0.031 | 200 | -0.038 <sup>o</sup> | 185 | +0.002 <sup>o</sup> | 93  |
| CLIP-L   | -0.017 | 200 | -0.025              | 200 | -0.045 | 199 | -0.031 | 200 | -0.038 <sup>o</sup> | 192 | -0.008 <sup>o</sup> | 124 |
| SigLIP-B | -0.013 | 200 | -0.025              | 200 | -0.053 | 200 | -0.027 | 200 | -0.052              | 199 | -0.044              | 197 |

(d) Backbones outside the ViT panel on the ImageNet centroid store (spectrum null, 20 replicates).

| model                    | $d$  | $\hat{\delta}$ | excess ( $z$ ) | reps above |
|--------------------------|------|----------------|----------------|------------|
| Barlow Twins (ResNet-50) | 2048 | 0.075          | -0.039 (-4.8)  | 20/20      |
| BYOL (ResNet-50)         | 2048 | 0.073          | -0.020 (-2.4)  | 20/20      |

Table 3: (continued) (c) Cosine reading, BH over 72 cells: sign-negative 70/72, genuine 56/72 and 22/24 on ImageNet+CIFAR-100, verdict agreement with the Euclidean record 65/72 cells. Text under the same cosine protocol (padding-free embeddings): genuine 8/15: gpt2-m, gpt2-l, gpt2-xl, pythia-410m, pythia-1b, pythia-2b8, olmo-1b, gte-qwen2. The angular tree: sibling-triplet agreement on CIFAR-100 for DINOv2-L is 0.91 under cosine against 0.71 under Euclidean distance. (d) Two self-supervised ResNet-50 backbones read the same clustered structure as the ViTs.

### A.3 IS THE READING ROBUST? QUADRUPLE BUDGET, NULL CONSTRUCTION, RESAMPLING AND CLASS COUNT

**Hierarchy depth, not class count.** Because  $C$  sets the quadruple population, we sweep ImageNet subsets  $C \in \{10, \dots, 1000\}$  in two compositions: random (spanning the full hierarchy) and WordNet-coherent (siblings, effectively *flat*). At fixed  $C$ , random subsets are more exploitable than coherent ones and carry a larger excess at small  $C$  (CLIP-L at every  $C$ , DINOv2-L up to  $C=100$ , DINOv2-G up to  $C=20$ ; beyond that the two compositions converge). Random subsets still score  $-0.08$  to  $-0.14$  at  $C=10$ , so MNIST’s flat verdict is not a class-count artifact, and DINOv2-G stays below its null at every  $C$  (excess  $-0.008$  at  $C=1000$ ,  $r=200$ ), where the supremum had placed it at the null.

(a) Quadruple budget: the excess at  $10^4$  to  $2 \times 10^6$  sampled quadruples per seed, under the record (Haar null, p99.9, 200 replicates) and under the superseded supremum  $\times$  Gaussian protocol.

| model    | data | protocol                   | $10^4$               | $5 \times 10^4$                                          | $10^5$  | $5 \times 10^5$ | $10^6$  | $2 \times 10^6$ | drift ( $\geq 10^5$ )            | drift/s.d. |
|----------|------|----------------------------|----------------------|----------------------------------------------------------|---------|-----------------|---------|-----------------|----------------------------------|------------|
| ViT-L    | IN   | record                     | -0.0161              | -0.0161                                                  | -0.0160 | -0.0157         | -0.0159 | -0.0159         | 0.0002                           | 0.07       |
|          |      | supremum, Gaussian         | -0.026               | -0.024                                                   | -0.027  | -0.030          | -0.034  | -0.035          | p99.9 @ $5 \times 10^5$ : -0.018 |            |
| DINOv2-L | IN   | record                     | -0.0131              | -0.0136                                                  | -0.0137 | -0.0135         | -0.0136 | -0.0136         | 0.0002                           | 0.38       |
|          |      | supremum, Gaussian         | -0.027               | -0.019                                                   | -0.024  | -0.011          | -0.004  | -0.005          | p99.9 @ $5 \times 10^5$ : -0.024 |            |
| CLIP-B   | IN   | record                     | -0.0063              | -0.0060                                                  | -0.0059 | -0.0065         | -0.0064 | -0.0064         | 0.0007                           | 0.13       |
|          |      | supremum, Gaussian         | -0.008               | -0.012                                                   | -0.023  | -0.025          | -0.025  | -0.031          | p99.9 @ $5 \times 10^5$ : -0.007 |            |
| ViT-L    | C100 | record                     | -0.0208              | -0.0208                                                  | -0.0206 | -0.0205         | -0.0205 | -0.0205         | 0.0002                           | 0.03       |
|          |      | supremum, Gaussian         | -0.045               | -0.047                                                   | -0.047  | -0.049          | -0.052  | -0.053          | p99.9 @ $5 \times 10^5$ : -0.033 |            |
| DINOv2-L | C100 | record                     | -0.0348              | -0.0348                                                  | -0.0348 | -0.0348         | -0.0347 | -0.0347         | 0.0001                           | 0.07       |
|          |      | supremum, Gaussian         | -0.071               | -0.055                                                   | -0.054  | -0.050          | -0.052  | -0.055          | p99.9 @ $5 \times 10^5$ : -0.069 |            |
| CLIP-B   | C100 | record                     | -0.0186              | -0.0195                                                  | -0.0188 | -0.0195         | -0.0194 | -0.0194         | 0.0006                           | 0.10       |
|          |      | supremum, Gaussian         | -0.045               | -0.047                                                   | -0.050  | -0.049          | -0.052  | -0.051          | p99.9 @ $5 \times 10^5$ : -0.030 |            |
| ViT-L    | DTD  | record                     | -0.0305              | -0.0308                                                  | -0.0312 | -0.0311         | -0.0313 | -0.0313         | 0.0002                           | 0.05       |
|          |      | supremum, Gaussian         | -0.064               | -0.070                                                   | -0.076  | -0.083          | -0.084  | -0.084          | p99.9 @ $5 \times 10^5$ : -0.054 |            |
| DINOv2-L | DTD  | record                     | -0.0418              | -0.0419                                                  | -0.0421 | -0.0420         | -0.0422 | -0.0420         | 0.0002                           | 0.06       |
|          |      | supremum, Gaussian         | -0.068               | -0.072                                                   | -0.075  | -0.083          | -0.084  | -0.084          | p99.9 @ $5 \times 10^5$ : -0.064 |            |
| CLIP-B   | DTD  | record                     | -0.0243              | -0.0235                                                  | -0.0242 | -0.0240         | -0.0243 | -0.0242         | 0.0003                           | 0.05       |
|          |      | supremum, Gaussian         | -0.048               | -0.055                                                   | -0.059  | -0.065          | -0.066  | -0.066          | p99.9 @ $5 \times 10^5$ : -0.035 |            |
| ViT-L    | C100 | centroid bootstrap: excess | -0.0493, s.d. 0.0029 | (30 resamples of the per-class images, 500 images/class) |         |                 |         |                 |                                  |            |
| DINOv2-L | C100 | centroid bootstrap: excess | -0.0493, s.d. 0.0022 | (30 resamples of the per-class images, 500 images/class) |         |                 |         |                 |                                  |            |
| DINOv2-G | C100 | centroid bootstrap: excess | -0.0778, s.d. 0.0020 | (30 resamples of the per-class images, 500 images/class) |         |                 |         |                 |                                  |            |
| CLIP-B   | C100 | centroid bootstrap: excess | -0.0476, s.d. 0.0017 | (30 resamples of the per-class images, 500 images/class) |         |                 |         |                 |                                  |            |
| ViT-L    | DTD  | centroid bootstrap: excess | -0.0803, s.d. 0.0049 | (30 resamples of the per-class images, 80 images/class)  |         |                 |         |                 |                                  |            |
| DINOv2-L | DTD  | centroid bootstrap: excess | -0.0878, s.d. 0.0048 | (30 resamples of the per-class images, 80 images/class)  |         |                 |         |                 |                                  |            |
| DINOv2-G | DTD  | centroid bootstrap: excess | -0.0863, s.d. 0.0051 | (30 resamples of the per-class images, 80 images/class)  |         |                 |         |                 |                                  |            |
| CLIP-B   | DTD  | centroid bootstrap: excess | -0.0581, s.d. 0.0054 | (30 resamples of the per-class images, 80 images/class)  |         |                 |         |                 |                                  |            |

(b) ImageNet centroid bootstrap under the record: 30 resamples of the 100 training images per class (20 null replicates per resample).

| model    | excess (reference) | bootstrap mean | bootstrap s.d. | fraction negative |
|----------|--------------------|----------------|----------------|-------------------|
| ViT-T    | -0.0055            | -0.0051        | 0.0011         | 1.00              |
| ViT-S    | -0.0126            | -0.0122        | 0.0007         | 1.00              |
| ViT-B    | -0.0165            | -0.0160        | 0.0006         | 1.00              |
| ViT-L    | -0.0160            | -0.0160        | 0.0004         | 1.00              |
| DINO-B   | -0.0209            | -0.0208        | 0.0003         | 1.00              |
| DINOv2-S | -0.0164            | -0.0162        | 0.0004         | 1.00              |
| DINOv2-B | -0.0167            | -0.0164        | 0.0002         | 1.00              |
| DINOv2-L | -0.0136            | -0.0134        | 0.0001         | 1.00              |
| DINOv2-G | -0.0080            | -0.0083        | 0.0004         | 1.00              |
| CLIP-B   | -0.0068            | -0.0069        | 0.0009         | 1.00              |
| CLIP-L   | -0.0094            | -0.0095        | 0.0006         | 1.00              |
| SigLIP-B | -0.0080            | -0.0081        | 0.0006         | 1.00              |

Table 4: **The excess is budget-stable, its size is partly a property of the null construction, and the genuine count survives resampling and estimator noise.** (a) Excess against the quadruple budget for nine cells; drift = range of the record excess over budgets  $\geq 10^5$ , divided by the excess’s spread in the last column. Under the record every sign holds at every budget and the drift of the excess from  $10^5$  upward is at most 0.38 of the excess’s own spread (null and estimator s.d. combined at  $5 \times 10^5$ ) on ImageNet, 0.10 on CIFAR-100 and 0.06 on DTD, so the excess is budget-stable; the supremum rows show the drift of up to 0.02 on ImageNet that the superseded protocol carried. (b) Bootstrap of the ImageNet excess over resampled images per class: the bootstrap s.d. is at most 0.0011 and every resample of every backbone is sign-negative.

(c) Three constructions of the spectrum-matched null (10 replicates each): Gaussian coefficients, Haar-rotated coefficients (exact sample spectrum), PC-permutation (exact per-component marginals).

| model    | IN     |        |          | C100   |        |          | C10    |        |          |
|----------|--------|--------|----------|--------|--------|----------|--------|--------|----------|
|          | Gauss. | Haar   | PC-perm. | Gauss. | Haar   | PC-perm. | Gauss. | Haar   | PC-perm. |
| ViT-T    | -0.005 | -0.014 | -0.038   | -0.044 | -0.031 | -0.055   | -0.057 | -0.027 | -0.050   |
| ViT-S    | -0.023 | -0.025 | -0.052   | -0.034 | -0.025 | -0.049   | -0.098 | -0.063 | -0.090   |
| ViT-B    | -0.027 | -0.025 | -0.054   | -0.018 | -0.011 | -0.039   | -0.090 | -0.054 | -0.078   |
| ViT-L    | -0.028 | -0.029 | -0.040   | -0.047 | -0.034 | -0.051   | -0.104 | -0.066 | -0.096   |
| DINO-B   | -0.005 | -0.002 | -0.008   | -0.068 | -0.056 | -0.068   | -0.100 | -0.070 | -0.097   |
| DINOv2-S | +0.010 | +0.015 | +0.013   | -0.050 | -0.029 | -0.051   | -0.093 | -0.083 | -0.098   |
| DINOv2-B | +0.004 | +0.015 | +0.006   | -0.042 | -0.007 | -0.041   | -0.132 | -0.120 | -0.124   |
| DINOv2-L | -0.013 | +0.000 | -0.011   | -0.053 | -0.008 | -0.044   | -0.153 | -0.136 | -0.122   |
| DINOv2-G | -0.000 | +0.016 | +0.001   | -0.080 | -0.032 | -0.074   | -0.155 | -0.138 | -0.110   |
| CLIP-B   | -0.028 | -0.028 | -0.048   | -0.046 | -0.036 | -0.053   | -0.079 | -0.045 | -0.075   |
| CLIP-L   | -0.016 | -0.017 | -0.029   | -0.042 | -0.031 | -0.052   | -0.087 | -0.049 | -0.082   |
| SigLIP-B | -0.019 | -0.020 | -0.044   | -0.042 | -0.032 | -0.046   | -0.096 | -0.056 | -0.081   |

(d) Joint sensitivity of the genuine count:  $z_{\text{joint}}$  against null, bootstrap and estimator noise, and (in parentheses) the number of the 30 centroid resamples in which the cell is genuine.

| model    | IN                     | C100                  | C10                    | DTD                    | FMNIST                  | MNIST                  |
|----------|------------------------|-----------------------|------------------------|------------------------|-------------------------|------------------------|
| ViT-T    | -1.1 (0) <sup>◦</sup>  | -0.7 (0) <sup>◦</sup> | -1.1 (0) <sup>◦</sup>  | -2.0 (13) <sup>†</sup> | +0.9 (0) <sup>◦</sup>   | -1.0 (0) <sup>◦</sup>  |
| ViT-S    | -3.6 (30)              | -0.0 (0) <sup>◦</sup> | -2.7 (30)              | -3.1 (30)              | -1.7 (30) <sup>†</sup>  | -1.3 (0) <sup>◦</sup>  |
| ViT-B    | -5.2 (30)              | +0.8 (0) <sup>◦</sup> | -2.4 (30)              | -4.0 (30)              | -2.1 (30)               | -0.6 (0) <sup>◦</sup>  |
| ViT-L    | -5.0 (30)              | -4.0 (30)             | -2.8 (30)              | -6.4 (30)              | -2.5 (30)               | -1.6 (0) <sup>◦</sup>  |
| DINO-B   | -8.9 (30)              | -7.6 (30)             | -2.7 (30)              | -4.5 (30)              | -0.3 (0) <sup>◦</sup>   | -1.5 (0) <sup>◦</sup>  |
| DINOv2-S | -15.9 (30)             | -9.7 (30)             | -2.8 (30)              | -3.2 (30)              | -1.6 (30) <sup>◦‡</sup> | -0.2 (0) <sup>◦</sup>  |
| DINOv2-B | -31.6 (30)             | -14.7 (30)            | -4.3 (30)              | -8.1 (30)              | -2.7 (30)               | -0.6 (0) <sup>◦</sup>  |
| DINOv2-L | -30.7 (30)             | -21.6 (30)            | -4.9 (30)              | -10.6 (30)             | -1.8 (29) <sup>†</sup>  | -0.5 (0) <sup>◦</sup>  |
| DINOv2-G | -11.5 (30)             | -28.5 (30)            | -4.8 (30)              | -8.2 (30)              | -2.0 (30)               | -0.8 (0) <sup>◦</sup>  |
| CLIP-B   | -1.3 (0) <sup>◦</sup>  | -3.1 (30)             | -1.8 (7) <sup>†</sup>  | -3.5 (30)              | -1.3 (0) <sup>◦</sup>   | -0.0 (0) <sup>◦</sup>  |
| CLIP-L   | -2.1 (12) <sup>†</sup> | -3.5 (30)             | -2.0 (30) <sup>†</sup> | -4.3 (30)              | -1.5 (0) <sup>◦</sup>   | -0.4 (0) <sup>◦</sup>  |
| SigLIP-B | -1.7 (0) <sup>◦</sup>  | -4.3 (30)             | -2.6 (30)              | -4.1 (30)              | -2.0 (30) <sup>†</sup>  | -1.9 (30) <sup>†</sup> |

Table 4: (continued) (c) Excess under three constructions of the spectrum null: Gaussian and PC-permutation agree, the exact-sample-spectrum construction yields smaller excesses for low-effective-rank clouds on the small- $C$  sets (most visibly DINOv2 on CIFAR-100), while signs and the ImageNet readings are stable. (d) Per cell:  $z_{\text{joint}} = \text{excess} / \sqrt{\sigma_{\text{null}}^2 + \sigma_{\text{boot}}^2 + \sigma_{\text{est}}^2}$  (200-replicate null s.d., bootstrap s.d. over 30 centroid resamples, quadruple-seed s.d.) and the number of resamples in which the cell is genuine under BH over the 72 cells (50 Haar replicates per resample); <sup>◦</sup>: not genuine in the record; <sup>†</sup>: record-genuine but failing one of the two criteria ( $z_{\text{joint}} \leq -2$ , genuine in  $\geq 27$  of 30); <sup>‡</sup>: not record-genuine but passing one. Counts: record 49/72 (18/24);  $z_{\text{joint}} \leq -2$ : 42/72 (18/24); genuine in  $\geq 27$  of 30 resamples: 47/72 (17/24).

(e) Class count against hierarchy depth: ImageNet subsets of  $C$  classes, random (spanning the hierarchy) or WordNet-coherent (siblings, effectively flat), under the record.

| model    | $C$  | random subsets (span the hierarchy) |        |       |        | WordNet-coherent (siblings) |        |       |        |
|----------|------|-------------------------------------|--------|-------|--------|-----------------------------|--------|-------|--------|
|          |      | $\hat{\delta}_{99.9}$               | excess | below | NC adv | $\hat{\delta}_{99.9}$       | excess | below | NC adv |
| DINOv2-L | 10   | 0.030                               | -0.134 | 5/5   | +1.04  | 0.068                       | -0.098 | 5/5   | +0.08  |
| DINOv2-L | 20   | 0.034                               | -0.091 | 5/5   | +1.32  | 0.071                       | -0.065 | 5/5   | +0.12  |
| DINOv2-L | 50   | 0.032                               | -0.046 | 5/5   | +1.58  | 0.050                       | -0.042 | 5/5   | +0.53  |
| DINOv2-L | 100  | 0.031                               | -0.032 | 5/5   | +1.46  | 0.044                       | -0.031 | 5/5   | +0.77  |
| DINOv2-L | 200  | 0.031                               | -0.022 | 5/5   | +1.45  | 0.041                       | -0.023 | 5/5   | +0.59  |
| DINOv2-L | 500  | 0.030                               | -0.016 | 2/2   | +1.01  | 0.031                       | -0.018 | 2/2   | +0.82  |
| DINOv2-L | 1000 | 0.030                               | -0.014 | 1/1   | +0.57  | —                           | —      | —     | —      |
| DINOv2-G | 10   | 0.029                               | -0.136 | 5/5   | +0.80  | 0.072                       | -0.095 | 5/5   | -0.56  |
| DINOv2-G | 20   | 0.037                               | -0.089 | 5/5   | +1.30  | 0.070                       | -0.067 | 5/5   | -0.18  |
| DINOv2-G | 50   | 0.039                               | -0.039 | 5/5   | +1.51  | 0.053                       | -0.042 | 5/5   | +0.26  |
| DINOv2-G | 100  | 0.041                               | -0.022 | 5/5   | +1.30  | 0.049                       | -0.028 | 5/5   | +0.36  |
| DINOv2-G | 200  | 0.036                               | -0.015 | 5/5   | +1.28  | 0.045                       | -0.020 | 5/5   | +0.27  |
| DINOv2-G | 500  | 0.034                               | -0.009 | 2/2   | +0.49  | 0.034                       | -0.013 | 2/2   | +0.22  |
| DINOv2-G | 1000 | 0.033                               | -0.008 | 1/1   | -0.13  | —                           | —      | —     | —      |
| CLIP-L   | 10   | 0.071                               | -0.080 | 5/5   | -0.08  | 0.099                       | -0.066 | 4/5   | -0.36  |
| CLIP-L   | 20   | 0.078                               | -0.047 | 5/5   | +0.20  | 0.113                       | -0.036 | 4/5   | -0.24  |
| CLIP-L   | 50   | 0.080                               | -0.027 | 5/5   | +0.34  | 0.113                       | -0.014 | 4/5   | -0.46  |
| CLIP-L   | 100  | 0.079                               | -0.024 | 5/5   | +0.21  | 0.095                       | -0.016 | 3/5   | -0.31  |
| CLIP-L   | 200  | 0.078                               | -0.019 | 5/5   | +0.00  | 0.086                       | -0.010 | 2/5   | -0.35  |
| CLIP-L   | 500  | 0.077                               | -0.012 | 2/2   | -0.27  | 0.081                       | -0.011 | 1/2   | -0.42  |
| CLIP-L   | 1000 | 0.076                               | -0.009 | 1/1   | -0.53  | —                           | —      | —     | —      |

Table 4: (continued) (e) Class-count control under the record (means over subset seeds: 5 for  $C \leq 200$ , 2 for  $C=500$ , 1 for  $C=1000$ ); “below” = subset seeds whose real value lies below the null at uncorrected  $p \leq 0.05$ ; NC adv = prototype-classifier advantage of the Poincaré readout in pp.

## A.4 DOES THE PREMISE SURVIVE AT THE SAMPLE LEVEL? THE CENSUS ON PER-IMAGE FEATURES

| model    | CIFAR-100 (10 images/class, $n=1000$ ) |                       |          |           |                 | DTD (22 images/class, $n=1034$ ) |                       |          |           |                 |
|----------|----------------------------------------|-----------------------|----------|-----------|-----------------|----------------------------------|-----------------------|----------|-----------|-----------------|
|          | sup. $\hat{\delta}$                    | $\hat{\delta}_{99.9}$ | excess   | exc./null | $r/200$ ( $p$ ) | sup. $\hat{\delta}$              | $\hat{\delta}_{99.9}$ | excess   | exc./null | $r/200$ ( $p$ ) |
| ViT-T    | 0.130                                  | 0.093                 | -0.0016° | -2%       | 134 (0.333)     | 0.124                            | 0.087                 | -0.0004° | -0%       | 111 (0.448)     |
| ViT-S    | 0.121                                  | 0.086                 | +0.0051° | +6%       | 21 (0.896)      | 0.103                            | 0.070                 | -0.0062  | -8%       | 197 (0.020)     |
| ViT-B    | 0.121                                  | 0.082                 | +0.0031° | +4%       | 50 (0.751)      | 0.092                            | 0.060                 | -0.0080  | -12%      | 200 (0.005)     |
| ViT-L    | 0.100                                  | 0.067                 | -0.0041° | -6%       | 185 (0.080)     | 0.092                            | 0.056                 | -0.0073  | -12%      | 200 (0.005)     |
| DINO-B   | 0.104                                  | 0.070                 | -0.0042° | -6%       | 194 (0.035)     | 0.110                            | 0.075                 | +0.0008° | +1%       | 80 (0.602)      |
| DINOv2-S | 0.100                                  | 0.065                 | -0.0049  | -7%       | 200 (0.005)     | 0.145                            | 0.090                 | +0.0058° | +7%       | 0 (1.000)       |
| DINOv2-B | 0.089                                  | 0.050                 | -0.0090  | -15%      | 200 (0.005)     | 0.139                            | 0.073                 | -0.0019° | -3%       | 162 (0.194)     |
| DINOv2-L | 0.087                                  | 0.043                 | -0.0131  | -24%      | 200 (0.005)     | 0.131                            | 0.064                 | -0.0060  | -9%       | 200 (0.005)     |
| DINOv2-G | 0.102                                  | 0.036                 | -0.0232  | -39%      | 200 (0.005)     | 0.136                            | 0.065                 | -0.0079  | -11%      | 200 (0.005)     |
| CLIP-B   | 0.117                                  | 0.080                 | -0.0060° | -7%       | 184 (0.085)     | 0.125                            | 0.089                 | +0.0021° | +2%       | 71 (0.647)      |
| CLIP-L   | 0.132                                  | 0.084                 | +0.0037° | +5%       | 35 (0.826)      | 0.120                            | 0.082                 | -0.0008° | -1%       | 124 (0.383)     |
| SigLIP-B | 0.113                                  | 0.078                 | -0.0074  | -9%       | 199 (0.010)     | 0.114                            | 0.079                 | -0.0044° | -5%       | 184 (0.085)     |

Table 5: **The sample-level reading, the object of prior latent-hyperbolicity claims, sits within null noise in most cells.** The instrument on  $\approx 1000$  stratified training images per cell (per-image features, not centroids): sup.  $\hat{\delta}$  is the raw supremum  $\delta_{\text{norm}}$  the literature reports;  $\hat{\delta}_{99.9}$ , excess, exc./null,  $r/200$  and  $p$  follow Table 1 (Haar spectrum-matched null on the sample cloud, 99.9th-percentile statistic, 200 replicates); °: not genuine under Benjamini–Hochberg over the 24 cells. The raw supremum lies in the band 0.087–0.145; the excess is not genuine in 14 of 24 cells and genuine in 10 (DINOv2-S, DINOv2-B, DINOv2-L, DINOv2-G, SigLIP-B on CIFAR-100; ViT-S, ViT-B, ViT-L, DINOv2-L, DINOv2-G on DTD), where it reaches at most -0.023, that is -39% to -7% of the null reading.

## A.5 IS THE STRUCTURE DEEPER THAN A STAR? THE DEPTH TEST

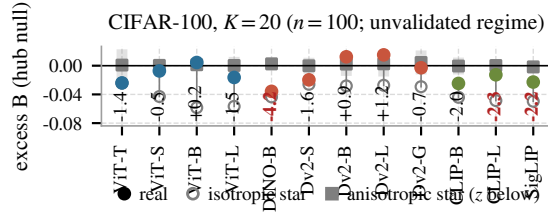


Figure 6: **The depth test on CIFAR-100 ( $K=20$ , five points per cluster) reads in the unvalidated regime.** Plotted: excess B under the hub-randomizing null for the real centroids (filled) and their matched stars (hollow circle: isotropic, withdrawn; square: anisotropic), with the  $z$  of real minus anisotropic star under each backbone.

**Real-data readings by regime.** ImageNet ( $n=1000$ , WordNet  $K=30$ ) lies in the validated regime: the real cloud is more hierarchical than its anisotropic matched star beyond the noise ( $z \leq -2$ ) in 4 of 12 backbones (ViT-S, ViT-B, ViT-L, DINOv2-L), never less, under Gaussian and under Haar-resampled star hubs alike (Table 6, Figure 4a). CIFAR-100 ( $n=100$ ,  $K=20$ ) lies in the unvalidated regime, where pure stars are declared more hierarchical in 17% of runs; its readings (3 of 12 with  $z \leq -2$ , none with  $z \geq +2$ ) are reported without certification (Figure 6). Validity domain of the test as run:  $n \approx 1000$ ; at  $n=100$  the false-alarm rate is 0% with  $K=6$  (17 points per cluster) and 17% with  $K \geq 12$ , so the test is not validated below about ten points per cluster. The bar was fixed before the sweep; the restriction to the validated regime was adopted after it.

**A trained control, inconclusive.** We fine-tuned the census ViT-B/16 for two passes over the images behind its ImageNet centroids with two objectives on the same batches: cross-entropy alone, and cross-entropy plus a hierarchical cross-entropy over the WordNet superclasses. Both fine-tuned models keep the census excess of the frozen backbone ( $-0.015$  and  $-0.016$  against  $-0.016$ ) and both are certified by the depth test, with  $z$  of  $-2.33$  and  $-2.77$  against  $-3.62$  for the frozen backbone: the hierarchical term lowers  $z$  relative to plain cross-entropy but neither reaches the frozen reading, so the control neither confirms nor refutes the test (Table 6). The full training set could not be read at training speed from the archive, so the schedule is two passes over the census subset.

(a) The twelve backbones: depth = excess B of the real centroids minus that of a matched star (negative = more hierarchical above the frame), with  $z$  against the combined spread.

| model    | C100, $K=20$ , Gaussian hubs |               | IN, $K=30$ , Gaussian hubs |               | IN, $K=30$ , Haar-resampled hubs |                 |                   | bal. frame |
|----------|------------------------------|---------------|----------------------------|---------------|----------------------------------|-----------------|-------------------|------------|
|          | iso. star                    | aniso. star   | iso. star                  | aniso. star   | star B                           | depth ( $z$ )   | $r_{\text{star}}$ | real $z$   |
| ViT-T    | +0.000 (+0.0)                | -0.025 (-1.4) | -0.015 (-1.5)              | -0.018 (-1.5) | +0.0011                          | -0.0184 (-1.45) | 0/10              | -2.1       |
| ViT-S    | +0.036 (+2.3)                | -0.008 (-0.5) | -0.021 (-1.9)              | -0.028 (-2.4) | +0.0010                          | -0.0257 (-2.04) | 0/10              | -1.7       |
| ViT-B    | +0.063 (+4.5)                | +0.003 (+0.2) | -0.026 (-2.9)              | -0.033 (-3.6) | +0.0012                          | -0.0347 (-3.01) | 0/10              | -2.8       |
| ViT-L    | +0.040 (+4.2)                | -0.017 (-1.5) | -0.024 (-3.2)              | -0.033 (-4.0) | +0.0000                          | -0.0337 (-3.73) | 0/10              | -5.6       |
| DINO-B   | +0.008 (+0.9)                | -0.038 (-4.2) | -0.002 (-0.2)              | -0.009 (-0.9) | +0.0002                          | -0.0092 (-1.02) | 0/10              | -1.0       |
| DINOv2-S | +0.006 (+0.6)                | -0.020 (-1.6) | +0.011 (+0.9)              | +0.008 (+0.7) | +0.0007                          | +0.0083 (+0.71) | 10/10             | +0.1       |
| DINOv2-B | +0.041 (+5.4)                | +0.010 (+0.9) | -0.011 (-0.9)              | -0.016 (-1.2) | +0.0018                          | -0.0155 (-1.18) | 0/10              | -0.3       |
| DINOv2-L | +0.042 (+6.8)                | +0.013 (+1.2) | -0.024 (-2.3)              | -0.028 (-2.4) | -0.0007                          | -0.0267 (-2.21) | 0/10              | -1.7       |
| DINOv2-G | +0.026 (+3.8)                | -0.007 (-0.7) | -0.024 (-1.6)              | -0.029 (-1.8) | -0.0021                          | -0.0263 (-1.50) | 0/10              | -0.9       |
| CLIP-B   | +0.019 (+1.8)                | -0.024 (-2.0) | -0.006 (-0.7)              | -0.016 (-2.0) | +0.0012                          | -0.0171 (-1.48) | 1/10              | -1.3       |
| CLIP-L   | +0.036 (+5.7)                | -0.013 (-2.3) | +0.004 (+0.5)              | -0.008 (-0.8) | +0.0018                          | -0.0090 (-0.74) | 2/10              | -0.7       |
| SigLIP-B | +0.027 (+2.8)                | -0.021 (-2.2) | +0.001 (+0.0)              | -0.011 (-0.9) | +0.0024                          | -0.0121 (-0.78) | 1/10              | -1.1       |

(b) Two ViT-B/16 supervised on ImageNet-1k leaf labels only, through the census protocol, against the IN-21k census backbone.

| model                          | IN excess ( $r$ ) | C100 excess ( $r$ ) | depth, Gaussian hubs ( $z$ ) | depth, Haar hubs ( $z$ ) |
|--------------------------------|-------------------|---------------------|------------------------------|--------------------------|
| DeiT-B, IN-1k                  | -0.0073 (200)     | -0.0257 (200)       | -0.0213 (-1.26)              | -0.0197 (-1.21)          |
| ViT-B, augreg, IN-1k           | -0.0070 (199)     | -0.0117 (187)       | -0.0249 (-3.03)              | -0.0290 (-2.65)          |
| ViT-B, augreg, IN-21k (census) | -0.0162 (200)     | +0.0057 (43)        | -0.0335 (-3.62)              | -0.0347 (-3.01)          |

Table 6: **Hierarchy above the superclasses is certified in four ImageNet backbones under both matched stars, imposing the geometry does not create it, and leaf-label supervision can produce it or not.** (a) Isotropic star: Gaussian clouds with each superclass’s RMS spread (the earlier, withdrawn construction); anisotropic star: within each superclass a Haar sample with the cloud’s own covariance, hubs Gaussian at the real hub radius; Haar-resampled hubs: the same clusters with the hubs drawn as a Haar resample of the real hubs. Hub-randomizing Haar null, 10 star seeds; frames: CIFAR-100 coarse labels, ImageNet WordNet cut. The test is validated at  $n=1000$  only (Table 7); CIFAR-100 readings ( $n=100$ ) are reported without certification. Certified on ImageNet ( $n=1000$ , the validated regime): 4 of 12 backbones (ViT-S, ViT-B, ViT-L, DINOv2-L) with  $z$  from  $-2.4$  to  $-4.0$ ; the withdrawn isotropic star reached  $z=+6.8$  on CIFAR-100. With the hubs of the matched star drawn as a Haar resample of the real hubs (exact hub spectrum, 10 star seeds) the certified set is ViT-S, ViT-B, ViT-L, DINOv2-L, with  $z$  from  $-2.04$  to  $-3.73$ ;  $r_{\text{star}}$  counts the star seeds whose excess B lies at or below the real one, a rank whose resolution is one part in eleven and which does not separate the certified backbones from the others. Balanced frame (complete-linkage WordNet cut with the smallest variance of cluster sizes, chosen before any depth run): certified ViT-T, ViT-B, ViT-L. K sweep: CIFAR-100  $K=5$ : iso 0/12 at  $z \leq -2$ , 7/12 at  $z \geq +2$ ; aniso 0/12 at  $z \leq -2$ , 0/12 at  $z \geq +2$ . CIFAR-100  $K=10$ : iso 0/12 at  $z \leq -2$ , 9/12 at  $z \geq +2$ ; aniso 1/12 at  $z \leq -2$ , 0/12 at  $z \geq +2$ . CIFAR-100  $K=20$ : iso 0/12 at  $z \leq -2$ , 8/12 at  $z \geq +2$ ; aniso 3/12 at  $z \leq -2$ , 0/12 at  $z \geq +2$ . ImageNet  $K=10$ : iso 0/12 at  $z \leq -2$ , 0/12 at  $z \geq +2$ ; aniso 2/12 at  $z \leq -2$ , 0/12 at  $z \geq +2$ . ImageNet  $K=30$ : iso 3/12 at  $z \leq -2$ , 0/12 at  $z \geq +2$ ; aniso 4/12 at  $z \leq -2$ , 0/12 at  $z \geq +2$ . ImageNet  $K=60$ : iso 3/12 at  $z \leq -2$ , 0/12 at  $z \geq +2$ ; aniso 4/12 at  $z \leq -2$ , 0/12 at  $z \geq +2$ . (b) Supervision on leaf labels can produce the depth (the augreg recipe is certified under both stars) and need not (DeiT-B is not detected): the hierarchical label set of IN-21k is not what makes the census ViTs deep; their WordNet alignment is in Table 10.

(c) A hyperbolic backbone and its Euclidean twin under the census of record and the depth test.

|            |         | CIFAR-100 ( $K=20$ , $n=100$ : depth unvalidated) |                                  |           | ImageNet ( $K=30$ , $n=1000$ : depth validated) |                                  |           |
|------------|---------|---------------------------------------------------|----------------------------------|-----------|-------------------------------------------------|----------------------------------|-----------|
| model      | input   | exc. ( $r$ )                                      | $\hat{\delta}_{99.9}$ nat./Eucl. | depth $z$ | exc. ( $r$ )                                    | $\hat{\delta}_{99.9}$ nat./Eucl. | depth $z$ |
| MERU ViT-S | images  | −0.020 (199)                                      | 0.117/0.117                      | −1.5      | −0.008° (186)                                   | 0.088/0.088                      | −1.5      |
| CLIP ViT-S | images  | −0.028 (200)                                      | 0.106/0.107                      | −3.9      | −0.006° (171)                                   | 0.110/0.088                      | −1.8      |
| MERU ViT-B | images  | −0.020 (200)                                      | 0.114/0.114                      | −2.2      | −0.010 (196)                                    | 0.087/0.087                      | −0.9      |
| CLIP ViT-B | images  | −0.030 (200)                                      | 0.105/0.103                      | −3.3      | −0.009° (187)                                   | 0.107/0.085                      | −1.0      |
| MERU ViT-L | images  | −0.012° (195)                                     | 0.118/0.118                      | −0.3      | −0.007° (184)                                   | 0.088/0.088                      | −0.5      |
| CLIP ViT-L | images  | −0.016 (199)                                      | 0.123/0.112                      | −1.7      | −0.010 (197)                                    | 0.104/0.084                      | −1.2      |
| MERU ViT-S | prompts | −0.022 (200)                                      | 0.086/0.086                      | −0.7      | −0.003° (155)                                   | 0.085/0.086                      | +0.0      |
| CLIP ViT-S | prompts | −0.023 (200)                                      | 0.085/0.082                      | −0.8      | +0.006° (15)                                    | 0.093/0.087                      | +0.8      |
| MERU ViT-B | prompts | −0.012 (200)                                      | 0.096/0.096                      | −0.7      | −0.001° (132)                                   | 0.089/0.090                      | +0.8      |
| CLIP ViT-B | prompts | −0.017 (200)                                      | 0.094/0.088                      | +0.3      | +0.008° (2)                                     | 0.100/0.092                      | +0.2      |
| MERU ViT-L | prompts | −0.006° (190)                                     | 0.102/0.102                      | +0.6      | +0.001° (75)                                    | 0.093/0.093                      | +0.5      |
| CLIP ViT-L | prompts | −0.012 (200)                                      | 0.098/0.092                      | +0.5      | +0.006° (5)                                     | 0.101/0.092                      | +0.4      |

(d) A trained control: the census ViT-B/16 fine-tuned with and without a hierarchical objective.

| ViT-B/16                                     | $\hat{\delta}_{99.9}$ | excess  | $r/200$ | depth   | $z$   |
|----------------------------------------------|-----------------------|---------|---------|---------|-------|
| frozen (census)                              | 0.054                 | −0.0162 | 200     | −0.0335 | −3.62 |
| cross-entropy                                | 0.048                 | −0.0153 | 200     | −0.0303 | −2.33 |
| cross-entropy + hierarchical CE (WordNet-30) | 0.048                 | −0.0162 | 200     | −0.0295 | −2.77 |

Table 6: (continued) (c) MERU (Desai et al., 2023) embeds images and text on the Lorentz hyperboloid with an entailment objective; its released twin is a CLIP baseline with the same ViT, RedCaps data and recipe minus the hyperbolic lift and entailment loss. Both pass through the instrument unchanged: class centroids of the projected embeddings (MERU: space-like hyperboloid coordinates; CLIP: unit vectors), Euclidean distances; exc. ( $r$ ): excess over the Haar spectrum-matched null with the number of 200 replicates above the real value (°: not genuine under BH over the 24 cells); nat./Eucl.: the same statistic in the model’s own metric (Lorentz distance between tangent-space-mean centroids for MERU, angular for CLIP) against the Euclidean one; depth  $z$ : the anisotropic matched-star test of panel (a) (10 star seeds), validated at  $n=1000$  only. On ImageNet images the excess runs from −0.007 to −0.010 for MERU and from −0.006 to −0.010 for its twin across ViT-S/B/L; MERU’s Lorentz-metric  $\hat{\delta}_{99.9}$  differs from its Euclidean value by at most 0.0002; depth  $z$  runs from −1.5 to −0.5 for MERU and from −1.8 to −1.0 for CLIP, so neither is more hierarchical than a matched star and MERU is never ahead of its twin. Its embeddings sit in the near-flat regime of the hyperboloid: learned curvature  $c=0.10$ , spatial norm times  $\sqrt{c}$  at the median 0.258–0.279 (95th percentile 0.263–0.287), and Lorentz-to-Euclidean pairwise distance ratio 0.9972–0.9991 at the median (0.9988–0.9997 on the class centroids), so the control tests the training objective, not a curved geometry. MERU’s hierarchy is a generic→specific (text⊃image) partial order, not a class taxonomy. (d) Two passes over the 100 training images per class behind the ImageNet centroids (patch embedding frozen, AdamW, same batches and seed for both objectives), with plain cross-entropy or with an added hierarchical cross-entropy over the WordNet 30-cut; both fine-tuned models stay certified and neither reaches the frozen reading, so the control is inconclusive.

#### A.6 HOW MUCH DEPTH CAN THE TEST SEE? POWER ON SYNTHETIC AND IMPLANTED HIERARCHIES

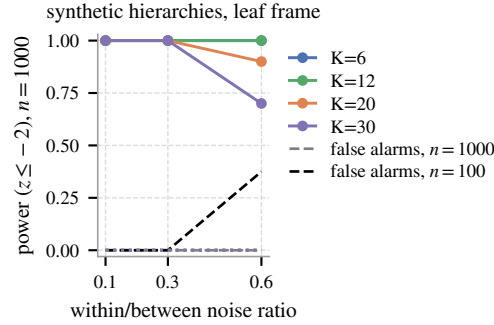


Figure 7: **The depth test has power on synthetic hierarchies at ImageNet’s class count when the within-cluster noise is below the between-hub spread.** Plotted: power ( $z \leq -2$ ) on synthetic two- and three-level hierarchies with the frame at the leaf clusters (solid, one line per  $K$ ) or at the top level (dotted), against the within/between noise ratio, with the false-alarm rates of pure stars dashed.

(a) Synthetic hierarchies at  $d=768$ : two- and three-level hierarchies (pooled) and pure stars, within/between noise ratios 0.1/0.3/0.6, isotropic and anisotropic clusters pooled, 5 seeds.

| frame                           | $n$  | $K$ | power @ ratio 0.1 | 0.3  | 0.6  | star $z \leq -2$ | star $z \geq +2$ |
|---------------------------------|------|-----|-------------------|------|------|------------------|------------------|
| top-level, isotropic star       | 100  | 6   | 0.00              | 0.00 | 0.00 | 0.00             | 0.17             |
| top-level, isotropic star       | 100  | 12  | 0.00              | 0.00 | 0.00 | 0.00             | 0.13             |
| top-level, isotropic star       | 100  | 20  | 0.00              | 0.00 | 0.00 | 0.00             | 0.17             |
| top-level, isotropic star       | 100  | 30  | 0.00              | 0.00 | 0.00 | 0.00             | 0.17             |
| top-level, isotropic star       | 1000 | 6   | 0.00              | 0.00 | 0.00 | 0.00             | 0.17             |
| top-level, isotropic star       | 1000 | 12  | 0.00              | 0.00 | 0.00 | 0.00             | 0.17             |
| top-level, isotropic star       | 1000 | 20  | 0.00              | 0.00 | 0.00 | 0.00             | 0.17             |
| top-level, isotropic star       | 1000 | 30  | 0.00              | 0.00 | 0.00 | 0.00             | 0.17             |
| leaf clusters, anisotropic star | 100  | 6   | 1.00              | 1.00 | 1.00 | 0.00             | 0.00             |
| leaf clusters, anisotropic star | 100  | 12  | 1.00              | 1.00 | 1.00 | 0.17             | 0.00             |
| leaf clusters, anisotropic star | 100  | 20  | 1.00              | 1.00 | 1.00 | 0.17             | 0.00             |
| leaf clusters, anisotropic star | 1000 | 6   | 1.00              | 1.00 | 1.00 | 0.00             | 0.00             |
| leaf clusters, anisotropic star | 1000 | 12  | 1.00              | 1.00 | 1.00 | 0.00             | 0.00             |
| leaf clusters, anisotropic star | 1000 | 20  | 1.00              | 1.00 | 0.90 | 0.00             | 0.00             |
| leaf clusters, anisotropic star | 1000 | 30  | 1.00              | 1.00 | 0.70 | 0.00             | 0.00             |

(b) Implanted two-level trees on the real ImageNet centroids: the hub arrangement replaced at strength  $s$ , every within-cluster offset kept; hits = implant seeds with  $z \leq -2$  at  $s=0/0.5/1$ ,  $\bar{z}_1$  = mean  $z$  at  $s=1$ , w/b = within/between spread.

| model    | WordNet-30 frame of record, Gaussian-hub star |     |       |       |             |                          | Haar-hub star |             | balanced frame (pre-specified) |       |       |
|----------|-----------------------------------------------|-----|-------|-------|-------------|--------------------------|---------------|-------------|--------------------------------|-------|-------|
|          | real $z$                                      | w/b | hits  | $s^*$ | $\bar{z}_1$ | tight $z$ at $s=0/0.5/1$ | hits $s=0/1$  | $\bar{z}_1$ | real $z$                       | hits  | $s^*$ |
| ViT-T    | -1.5                                          | 1.5 | 0/0/0 | -     | +0.1        | -0.6/-1.9/-9.2           | 0/0           | +0.0        | -2.1                           | 0/0/1 | -     |
| ViT-S    | -2.4                                          | 1.9 | 0/0/0 | -     | -1.2        | -0.2/-3.4/-10.1          | 0/0           | -0.7        | -1.7                           | 0/0/0 | -     |
| ViT-B    | -3.6                                          | 2.0 | 0/0/0 | -     | +0.4        | -1.0/-3.9/-11.5          | 0/0           | +0.4        | -2.8                           | 0/0/0 | -     |
| ViT-L    | -4.0                                          | 1.9 | 0/0/0 | -     | -1.4        | -1.6/-4.4/-12.7          | 0/0           | -1.0        | -5.6                           | 0/0/0 | -     |
| DINO-B   | -0.9                                          | 2.1 | 0/0/0 | -     | -0.3        | -1.6/-5.1/-12.1          | 0/0           | -0.3        | -1.0                           | 0/0/0 | -     |
| DINOv2-S | +0.7                                          | 3.0 | 0/0/0 | -     | -0.8        | -0.2/-3.1/-11.5          | 0/0           | -0.8        | +0.1                           | 0/0/0 | -     |
| DINOv2-B | -1.2                                          | 3.7 | 0/0/0 | -     | +0.4        | -0.0/-3.1/-11.2          | 0/0           | +0.5        | -0.3                           | 0/0/0 | -     |
| DINOv2-L | -2.4                                          | 3.9 | 0/0/0 | -     | +0.2        | +0.2/-2.7/-11.4          | 0/0           | +0.2        | -1.7                           | 0/0/0 | -     |
| DINOv2-G | -1.8                                          | 3.9 | 0/0/0 | -     | +1.3        | +0.2/-3.9/-11.6          | 0/0           | +1.2        | -0.9                           | 0/0/0 | -     |
| CLIP-B   | -2.0                                          | 1.3 | 0/0/0 | -     | -1.2        | -1.7/-3.8/-11.4          | 0/0           | -0.7        | -1.3                           | 0/0/0 | -     |
| CLIP-L   | -0.8                                          | 1.5 | 0/0/3 | -     | -2.2        | -2.4/-5.8/-13.5          | 0/1           | -1.7        | -0.7                           | 0/0/0 | -     |
| SigLIP-B | -0.9                                          | 1.5 | 0/0/0 | -     | -1.6        | -2.1/-5.0/-11.0          | 0/0           | -1.1        | -1.1                           | 0/0/0 | -     |

Table 7: **The depth test meets its bar on synthetic hierarchies at ImageNet’s class count, raises no false alarm on real clouds under either star, and misses implanted depth at the real within-cluster spread.** (a) Power = fraction of hierarchy runs with  $z \leq -2$ ; star columns = false-alarm rates in each direction. Top-level frame: the test receives the  $K$  super-cluster labels, as the earlier isotropic test did on the real backbones; its hub null keeps every frame cluster intact and only rearranges the  $K$  hubs, so hierarchy below the frame is invisible by construction. Leaf frame: the test receives the finest cluster labels with the anisotropic matched star (configurations with more leaves than points are infeasible and omitted). Pre-set bar: power  $\geq 0.8$  for two- and three-level hierarchies at noise ratios  $\leq 0.3$  for both  $n$ , and false alarms  $\leq 5\%$  in each direction. Leaf frame: met at  $n=1000$  (power 1.00 at ratios  $\leq 0.3$  and 0.90 at 0.6; false alarms 0% and 0%), failed at  $n=100$ , where  $K \geq 12$  (under ten points per cluster) gives 17% false alarms at  $z \leq -2$ . Top-level frame: power 0.00, false alarms 16% pooled. (b) Six super-hubs of five, hub =  $s$ -super-hub +  $(1 - s + 0.4s)$ -own Gaussian draw, hub cloud rescaled to the real RMS radius; the depth test of Table 6 runs unchanged, 5 implant seeds per  $s$ . hits: implant seeds with  $z \leq -2$  (of 5);  $s^*$ : first  $s$  with  $\geq 4$  of 5; within/between: RMS of the within-cluster offsets over the point-weighted RMS of the hub displacements (1.3 to 3.9 on the real clouds, against 0.1–0.6 in the synthetic sweep); tight: the same implant after shrinking the offsets to within/between = 0.6 (2 seeds). Power (fraction of runs with  $z \leq -2$ ) by  $s$ :  $s=0$ : 0.00,  $s=0.25$ : 0.00,  $s=0.5$ : 0.00,  $s=0.75$ : 0.00,  $s=1$ : 0.05; false alarms at  $s=0$ : 0 of 60. Under the Haar-hub star the same implants give 0 of 60 false alarms at  $s=0$  and 1 of 60 detections at  $s=1$  (power 0.02). Balanced frame: the  $K=30$  agglomerative cut of the WordNet distance matrix (complete linkage) with the smallest variance of cluster sizes, chosen and logged before any depth run; sizes 1–154 classes against 1–316 for the frame of record.

#### A.7 IS THE STRUCTURE LEARNED? TRAINING INTERVENTIONS AND CURVATURE BY EDGE TYPE

**Clustered structure tracks training.** Two interventions move it (Figure 8; raw  $\delta$ , within architecture): fine-tuning on a task with no class hierarchy raises  $\delta$  by 19–91%, and  $\delta$  falls across transformer depth. Shuffling the embedding-to-label correspondence leaves it unchanged, as a null-invariant quantity should. Random-weight networks give higher raw  $\delta$  but near-Gaussian features, so they are not counted as evidence; excess-based versions of these interventions await stored point clouds. On matched architecture (ViT-B supervised, DINO and CLIP) all three objectives yield a genuine excess on most datasets, and the family separations emerge with scale rather than at fixed size. High mean Ollivier–Ricci curvature coexists with tree-likeness because they live on different edges: negative curvature concentrates on between-superclass bridges, about twice as often as within superclasses on ImageNet, while the mean measures within-cluster compactness (Table 8).

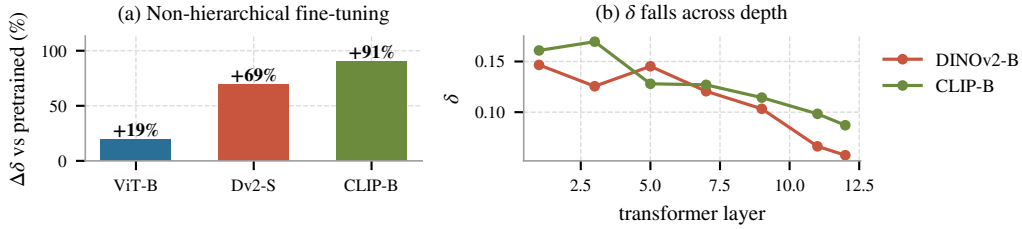


Figure 8: **The raw reading is learned: training interventions move it.** Plotted: raw  $\delta$  within architecture, (a) after non-hierarchical fine-tuning against the pretrained value, and (b) across transformer depth (layers  $\geq 1$ ).

(a) Interventions on the raw reading, within architecture.

| intervention                                     | model    | $\delta$ before | $\delta$ after | change |
|--------------------------------------------------|----------|-----------------|----------------|--------|
| non-hierarchical fine-tuning                     | DINOv2-S | 0.095           | 0.161          | +69%   |
| non-hierarchical fine-tuning                     | ViT-B    | 0.097           | 0.116          | +19%   |
| non-hierarchical fine-tuning                     | CLIP-B   | 0.104           | 0.199          | +91%   |
| across depth (layer 1 $\rightarrow$ 12)          | DINOv2-B | 0.147           | 0.058          | −61%   |
| across depth (layer 1 $\rightarrow$ 12)          | CLIP-B   | 0.161           | 0.087          | −46%   |
| random weights (pretrained $\rightarrow$ random) | DINOv2-S | 0.095           | 0.222          | +134%  |
| random weights (pretrained $\rightarrow$ random) | DINOv2-B | 0.081           | 0.198          | +144%  |
| random weights (pretrained $\rightarrow$ random) | ViT-B    | 0.113           | 0.187          | +65%   |

(b) Matched architecture, three objectives: excess of ViT-B/16 supervised, DINO and CLIP.

| dataset   | ViT-B (supervised) | DINO-B | CLIP-B |
|-----------|--------------------|--------|--------|
| ImageNet  | −0.027             | −0.004 | −0.030 |
| CIFAR-100 | −0.019             | −0.072 | −0.050 |
| CIFAR-10  | −0.081             | −0.090 | −0.069 |
| DTD       | −0.064             | −0.053 | −0.065 |
| FMNIST    | −0.071             | −0.037 | −0.051 |
| MNIST     | −0.044             | −0.069 | −0.036 |

(c) Ollivier–Ricci curvature by edge type on the centroid kNN graph (30 WordNet superclasses).

| model    | dataset | ORC within | ORC across | %neg within | %neg across |
|----------|---------|------------|------------|-------------|-------------|
| ViT-T    | IN      | 0.35       | 0.27       | 5.2         | 9.4         |
| ViT-T    | C100    | 0.51       | 0.42       | 1.4         | 2.7         |
| ViT-S    | IN      | 0.37       | 0.28       | 3.4         | 8.3         |
| ViT-S    | C100    | 0.61       | 0.48       | 0.0         | 3.0         |
| ViT-B    | IN      | 0.40       | 0.30       | 2.1         | 5.9         |
| ViT-B    | C100    | 0.61       | 0.48       | 0.0         | 1.7         |
| ViT-L    | IN      | 0.40       | 0.31       | 2.1         | 6.8         |
| ViT-L    | C100    | 0.61       | 0.47       | 0.6         | 2.1         |
| DINO-B   | IN      | 0.40       | 0.33       | 2.8         | 4.1         |
| DINO-B   | C100    | 0.60       | 0.45       | 0.0         | 1.4         |
| DINOv2-S | IN      | 0.37       | 0.33       | 1.8         | 1.2         |
| DINOv2-S | C100    | 0.60       | 0.44       | 0.0         | 0.8         |
| DINOv2-B | IN      | 0.58       | 0.64       | 0.3         | 0.1         |
| DINOv2-B | C100    | 0.66       | 0.54       | 0.0         | 0.1         |
| DINOv2-L | IN      | 0.69       | 0.72       | 0.0         | 0.0         |
| DINOv2-L | C100    | 0.70       | 0.64       | 0.0         | 0.1         |
| DINOv2-G | IN      | 0.70       | 0.74       | 0.0         | 0.0         |
| DINOv2-G | C100    | 0.77       | 0.70       | 0.0         | 0.0         |
| CLIP-B   | IN      | 0.36       | 0.29       | 4.2         | 7.9         |
| CLIP-B   | C100    | 0.60       | 0.46       | 0.0         | 3.1         |
| CLIP-L   | IN      | 0.35       | 0.27       | 4.0         | 8.4         |
| CLIP-L   | C100    | 0.57       | 0.43       | 0.0         | 2.2         |
| SigLIP-B | IN      | 0.34       | 0.26       | 5.0         | 7.5         |
| SigLIP-B | C100    | 0.59       | 0.43       | 0.0         | 2.3         |

Table 8: **The raw reading is learned, and negative curvature lives on the bridges between superclasses.** (a) Fine-tuning on a task with no class hierarchy raises  $\delta$ ,  $\delta$  falls across transformer depth, and random-weight networks read a higher raw  $\delta$  with near-Gaussian features, so they are not counted as evidence; label shuffling leaves  $\delta$  unchanged, as a null-invariant quantity should (Figure 8). (b) The three objectives on one architecture; the family separations emerge with scale rather than at fixed size. (c) Negative curvature concentrates on between-cluster bridges while the mean ORC measures within-cluster compactness; DINOv2 on ImageNet inverts the pattern and recovers it on CIFAR-100.

## A.8 IS THE ISLAND REAL? THE TREE MAP UNDER SIX CONFIGURATIONS

Threshold sensitivity of the selection criterion of §6.4: the selected configuration is stable for degeneracy bounds in  $[0.45, 0.50]$  on ImageNet and  $[0.40, 0.50]$  on CIFAR-100; tighter bounds select cosine-complete on ImageNet (gap 0.13) and cosine-Ward on CIFAR-100 (gap 0.56), and in every case the label-free gap is moderate and configuration-sensitive, never a categorical island. The criterion must be applied per dataset: the configuration it selects on DBpedia (Euclidean, average linkage, where nothing is degenerate) is exactly the degenerate naive configuration on ImageNet, so a criterion transferred across datasets would reintroduce the island.

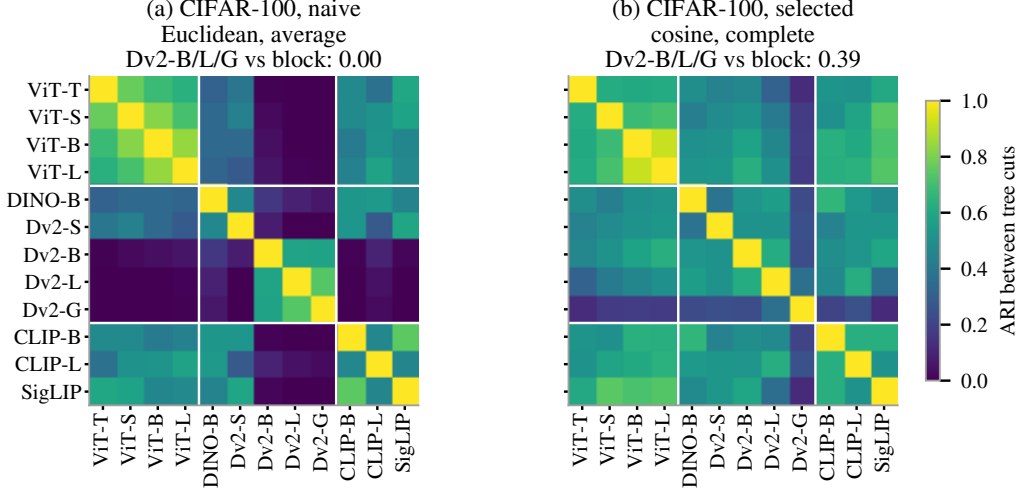


Figure 9: **The island is an artifact on CIFAR-100 too.** Plotted: the same reading as Figure 5 on the 20 CIFAR-100 superclass cuts, naive (a) against criterion-selected (b).

(a) Selection diagnostics and mean pairwise ARI at the reference cut.

| dataset   | configuration              | max frac. | CPCC  | mean pairwise ARI at the reference cut |                    |        |
|-----------|----------------------------|-----------|-------|----------------------------------------|--------------------|--------|
|           |                            |           |       | B/L/G vs block                         | S, DINO-B vs block | within |
| ImageNet  | euclid-average (deg.)      | 0.97      | 0.761 | 0.03                                   | 0.08               | 0.64   |
| ImageNet  | euclid-complete (deg.)     | 0.83      | 0.606 | 0.03                                   | 0.17               | 0.44   |
| ImageNet  | euclid-ward                | 0.48      | 0.464 | 0.15                                   | 0.38               | 0.53   |
| ImageNet  | cosine-average (selected)  | 0.40      | 0.691 | 0.38                                   | 0.35               | 0.48   |
| ImageNet  | cosine-complete            | 0.32      | 0.539 | 0.13                                   | 0.17               | 0.50   |
| ImageNet  | cosine-ward                | 0.37      | 0.464 | 0.17                                   | 0.39               | 0.55   |
| CIFAR-100 | euclid-average (deg.)      | 0.80      | 0.798 | 0.00                                   | 0.41               | 0.57   |
| CIFAR-100 | euclid-complete (deg.)     | 0.66      | 0.722 | 0.17                                   | 0.40               | 0.54   |
| CIFAR-100 | euclid-ward                | 0.33      | 0.534 | 0.37                                   | 0.65               | 0.63   |
| CIFAR-100 | cosine-average (deg.)      | 0.53      | 0.777 | 0.30                                   | 0.54               | 0.58   |
| CIFAR-100 | cosine-complete (selected) | 0.39      | 0.698 | 0.39                                   | 0.51               | 0.63   |
| CIFAR-100 | cosine-ward                | 0.14      | 0.584 | 0.56                                   | 0.62               | 0.61   |

(b) Three measures of DINOv2-B/L/G against the supervised+contrastive block, each written as vs block / within block.

| dataset   | configuration              | ARI at the cut | cophenetic corr. | triplet agreement |
|-----------|----------------------------|----------------|------------------|-------------------|
| ImageNet  | euclid-average (deg.)      | 0.03 / 0.64    | 0.36 / 0.80      | 0.47 / 0.74       |
| ImageNet  | euclid-complete (deg.)     | 0.03 / 0.44    | 0.20 / 0.66      | 0.41 / 0.65       |
| ImageNet  | euclid-ward                | 0.16 / 0.53    | 0.62 / 0.84      | 0.67 / 0.85       |
| ImageNet  | cosine-average (selected)  | 0.39 / 0.48    | 0.48 / 0.78      | 0.77 / 0.76       |
| ImageNet  | cosine-complete            | 0.13 / 0.49    | 0.18 / 0.72      | 0.36 / 0.71       |
| ImageNet  | cosine-ward                | 0.17 / 0.55    | 0.59 / 0.85      | 0.69 / 0.86       |
| CIFAR-100 | euclid-average (deg.)      | 0.00 / 0.57    | 0.15 / 0.78      | 0.41 / 0.74       |
| CIFAR-100 | euclid-complete (deg.)     | 0.17 / 0.54    | 0.23 / 0.72      | 0.41 / 0.66       |
| CIFAR-100 | euclid-ward                | 0.37 / 0.63    | 0.63 / 0.77      | 0.70 / 0.79       |
| CIFAR-100 | cosine-average (deg.)      | 0.30 / 0.58    | 0.42 / 0.72      | 0.51 / 0.64       |
| CIFAR-100 | cosine-complete (selected) | 0.39 / 0.63    | 0.46 / 0.64      | 0.50 / 0.61       |
| CIFAR-100 | cosine-ward                | 0.56 / 0.61    | 0.74 / 0.73      | 0.76 / 0.76       |

Table 9: **The categorical island is a chaining artifact of the naive cut; a moderate gap remains under every admissible configuration and closes only in triplet agreement under cosine-average.** (a) Six configurations (metric  $\times$  linkage) per dataset with the two quantities that drive the selection criterion of §6.4: the degeneracy diagnostic (largest-cluster fraction at the reference cut, worst model) and the mean cophenetic fidelity (CPCC) of each model’s dendrogram to its own distance matrix; configurations with max-cluster fraction  $> 0.5$  are excluded and, among the rest, the highest CPCC selects the marked row; then the mean pairwise ARI between the cuts at 30 (ImageNet) or 20 (CIFAR-100) clusters. (b) Dendrograms rebuilt from the census cache: ARI at the cut, Pearson correlation between the two trees’ cophenetic distance vectors, and agreement on which pair merges first over  $10^4$  random class triplets. Under the naive ImageNet configuration, cross-family ARI between the supervised ViTs and the contrastive VLMs reaches 0.82 while every DINOv2-vs-block pair stays at or below 0.06, and the worst model’s largest cluster holds 97% of the classes. Under the selected configuration the DINOv2 family agrees with the block at 0.38 against 0.48 within the block. Under the other 3 admissible ImageNet configurations the DINOv2-vs-block agreement is 0.13–0.17 against 0.50–0.55 within the block; over the admissible configurations and the three measures the median fraction of within-block agreement missing is 0.34, about a third.

#### A.9 WHOSE TAXONOMY? WORDNET ALIGNMENT, SUPERCLASS RECOVERY, DBPEDIA AND HIERARCAPS

On HierarCaps (Alper & Averbuch-Elor, 2024) (1000 four-level caption chains), text encoders show the same reading with no class set: the projection radius orders levels within a chain, sibling triplets are resolved at 0.83–0.97 (chance 0.5), and the leaf excess repeats the text census (Table 10).

| (a) Alignment with WordNet on ImageNet.                                                                                                                                                              |                                                      |             |                       |                  |                                               |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|-------------|-----------------------|------------------|-----------------------------------------------|
| model                                                                                                                                                                                                | Spearman $\rho$ , inter-centroid vs WordNet distance |             |                       |                  | raw $\hat{\delta}$ of WordNet / random groups |
|                                                                                                                                                                                                      | $\rho_{WN}$                                          | shuffle     | Gaussian              | single image     |                                               |
| ViT-T                                                                                                                                                                                                | +0.52                                                | +0.02       | +0.01                 | –                | 0.146 / 0.159                                 |
| ViT-S                                                                                                                                                                                                | +0.50                                                | +0.00       | –0.00                 | –                | 0.110 / 0.130                                 |
| ViT-B                                                                                                                                                                                                | +0.49                                                | +0.03       | +0.01                 | –                | 0.092 / 0.120                                 |
| ViT-L                                                                                                                                                                                                | +0.53                                                | +0.01       | +0.03                 | +0.25            | 0.104 / 0.117                                 |
| DINO-B                                                                                                                                                                                               | +0.36                                                | –0.01       | –0.01                 | –                | 0.084 / 0.117                                 |
| DINOv2-S                                                                                                                                                                                             | +0.22                                                | +0.01       | +0.01                 | –                | 0.077 / 0.100                                 |
| DINOv2-B                                                                                                                                                                                             | +0.18                                                | –0.01       | +0.01                 | –                | 0.068 / 0.069                                 |
| DINOv2-L                                                                                                                                                                                             | +0.18                                                | –0.00       | –0.03                 | +0.08            | 0.060 / 0.058                                 |
| DINOv2-G                                                                                                                                                                                             | +0.20                                                | –0.03       | –0.01                 | +0.03            | 0.054 / 0.056                                 |
| CLIP-B                                                                                                                                                                                               | +0.57                                                | +0.00       | +0.01                 | –                | 0.126 / 0.174                                 |
| CLIP-L                                                                                                                                                                                               | +0.59                                                | +0.01       | +0.03                 | +0.26            | 0.117 / 0.162                                 |
| SigLIP-B                                                                                                                                                                                             | +0.58                                                | –0.01       | +0.01                 | –                | 0.118 / 0.158                                 |
| DeiT-B (IN-1k)                                                                                                                                                                                       | +0.08                                                | +0.01       | –                     | –                | –                                             |
| ViT-B (augreg, IN-1k)                                                                                                                                                                                | +0.52                                                | +0.01       | –                     | –                | –                                             |
| (b) Recovery of the CIFAR-100 superclasses: ARI of the 20-cluster cut per configuration ( $\dagger$ : degenerate on CIFAR-100, euclid-average being the naive configuration; *: criterion-selected). |                                                      |             |                       |                  |                                               |
| model                                                                                                                                                                                                | euclid-avg $^\dagger$                                | euclid-ward | cosine-avg $^\dagger$ | cosine-complete* | cosine-ward                                   |
| ViT-T                                                                                                                                                                                                | 0.31                                                 | 0.38        | 0.29                  | 0.33             | 0.33                                          |
| ViT-S                                                                                                                                                                                                | 0.35                                                 | 0.56        | 0.37                  | 0.47             | 0.56                                          |
| ViT-B                                                                                                                                                                                                | 0.39                                                 | 0.61        | 0.43                  | 0.45             | 0.57                                          |
| ViT-L                                                                                                                                                                                                | 0.38                                                 | 0.55        | 0.49                  | 0.48             | 0.60                                          |
| DINO-B                                                                                                                                                                                               | 0.22                                                 | 0.48        | 0.28                  | 0.38             | 0.54                                          |
| DINOv2-S                                                                                                                                                                                             | 0.12                                                 | 0.51        | 0.32                  | 0.33             | 0.49                                          |
| DINOv2-B                                                                                                                                                                                             | 0.02                                                 | 0.47        | 0.35                  | 0.42             | 0.51                                          |
| DINOv2-L                                                                                                                                                                                             | 0.01                                                 | 0.32        | 0.25                  | 0.50             | 0.48                                          |
| DINOv2-G                                                                                                                                                                                             | 0.00                                                 | 0.19        | 0.08                  | 0.20             | 0.52                                          |
| CLIP-B                                                                                                                                                                                               | 0.18                                                 | 0.48        | 0.18                  | 0.37             | 0.48                                          |
| CLIP-L                                                                                                                                                                                               | 0.35                                                 | 0.52        | 0.32                  | 0.41             | 0.54                                          |
| SigLIP-B                                                                                                                                                                                             | 0.22                                                 | 0.49        | 0.22                  | 0.37             | 0.52                                          |
| DeiT-B (IN-1k)                                                                                                                                                                                       | 0.22                                                 | 0.56        | 0.23                  | 0.38             | 0.54                                          |
| ViT-B (augreg, IN-1k)                                                                                                                                                                                | 0.30                                                 | 0.44        | 0.40                  | 0.42             | 0.48                                          |

Table 10: **Alignment with the human taxonomy follows supervision and recipe, the superclasses are recovered by every family, and the recovery is not WordNet circularity.** (a)  $\rho_{WN}$ : Spearman correlation between inter-centroid and WordNet tree distances on ImageNet, with the label-shuffle and Gaussian-cloud controls; single image: the same correlation on single-image distances (one image per class); WordNet / random groups: raw  $\hat{\delta}$  of WordNet-coherent against random class groupings. (b) ARI of the 20-cluster cut against the true CIFAR-100 superclasses per configuration (degenerate configurations marked; the criterion-selected one is cosine-complete). Paired within each of the 3 admissible configurations, the label-supervised ViT-B recovers more than each DINOv2 model in 11 of 12 comparisons; DINOv2 itself recovers the superclasses at ARI 0.19–0.52, and the best cut over models and configurations reaches 0.61 against zero for permuted labels. Circularity controls: WordNet-coherent groupings read a lower raw  $\delta$  than random ones in 11 of 12 models, and single-image distances still correlate with WordNet at up to +0.26. The two leaf-label ViT-B/16 of Table 6: the augreg recipe is aligned with WordNet like the IN-21k ViTs ( $\rho_{WN}$  +0.52) while DeiT-B is nearly unaligned (+0.08) yet recovers the CIFAR-100 superclasses at ARI up to 0.56, so WordNet alignment is recipe-dependent even among leaf-supervised ViTs.

(c) DBpedia Classes (219 leaf classes, 3 levels), a text hierarchy independent of WordNet; gains in pp.

| model    | supremum, Gaussian    |        | census of record      |        |             | trip. | NC H-R | FS H-R     |
|----------|-----------------------|--------|-----------------------|--------|-------------|-------|--------|------------|
|          | $\hat{\delta}_{\max}$ | excess | $\hat{\delta}_{99.9}$ | excess | $r(p)$      |       |        |            |
| BGE-base | 0.122                 | -0.028 | 0.082                 | -0.021 | 200 (0.005) | 0.89  | -0.59  | +0.05±0.05 |
| E5-base  | 0.127                 | -0.024 | 0.086                 | -0.017 | 200 (0.005) | 0.89  | -0.43  | +0.04±0.03 |
| GTE-base | 0.132                 | -0.020 | 0.088                 | -0.017 | 200 (0.005) | 0.89  | -0.82  | +0.08±0.06 |

(d) HierarCaps (1000 four-level caption chains): a multi-level text hierarchy with no class set.

| model    | $\rho(\text{level}, \text{radius})$ | % monotone | trip. L1 | trip. L2 | $\hat{\delta}$ leaves | excess |
|----------|-------------------------------------|------------|----------|----------|-----------------------|--------|
| BGE-base | +0.80±0.31                          | 49.6       | 0.88     | 0.97     | 0.130                 | +0.003 |
| E5-base  | +0.87±0.19                          | 53.9       | 0.90     | 0.97     | 0.139                 | +0.010 |
| GTE-base | +0.47±0.55                          | 24.0       | 0.90     | 0.97     | 0.135                 | +0.006 |
| CLIP-B   | +0.64±0.43                          | 27.2       | 0.86     | 0.95     | 0.123                 | -0.011 |
| CLIP-L   | +0.51±0.49                          | 18.7       | 0.86     | 0.96     | 0.117                 | -0.010 |
| SigLIP-B | +0.32±0.55                          | 9.8        | 0.83     | 0.91     | 0.138                 | +0.014 |

Table 10: (continued) (c) Excess under the original supremum reading (3 Gaussian replicates) and under the census of record (Haar null, 99.9th-percentile statistic, 200 replicates;  $r$  = replicates above the real value, left-tail  $p$ ), sibling-triplet agreement, and the metric gains, marginal as the diagnostic predicts. Tree map on the same classes: no configuration degenerates (largest cluster  $\leq 10\%$ ), the embedders recover the true 70-way level-2 partition at ARI 0.21–0.29 in all 6 configurations and agree with each other at cross-model ARI 0.65–0.82; the configuration the criterion selects there (Euclidean, average linkage) is the one that degenerates on ImageNet, which is why the criterion is applied per dataset. (d) Radial ordering of the four levels under the paper’s projection (chance for strict monotonicity =  $1/24 \approx 4.2\%$ ; shuffle controls  $\approx 0$ ), sibling triplet agreement (cosine; chance 0.5), and leaf  $\hat{\delta}$  with spectrum-null excess.

#### A.10 IS TEXT THE SAME? THE TEXT CENSUS UNDER THE RECORD

**Extraction, anisotropy and templates.** Extraction is part of the measurement: left-padded batching moves GPT-2’s  $\hat{\delta}$  by up to 0.018, of the order of the fragile excesses; the padding-free protocol is the record. Genuineness is not a proxy for anisotropy: GTE-base has the highest effective rank and is not genuine, and the only LLM-backed embedder tested (GTE-Qwen2-1.5B) is not genuine under the record. Causal-LM values are template-sensitive but the scale ordering is template-robust (Table 11).

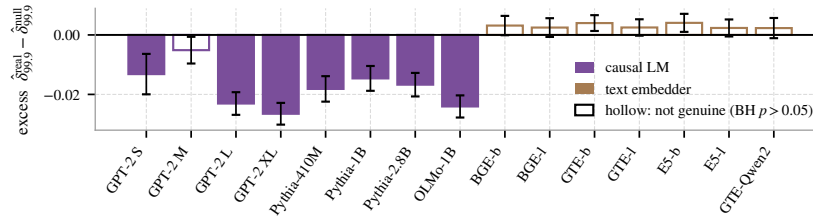


Figure 10: **In text, clustered structure depends on recipe and scale: robust at the larger GPT-2 sizes, fragile at the smaller.** Plotted: the text census of record (padding-free extraction, Haar null, 99.9th-percentile statistic, 200 replicates), with error bars combining null and estimator s.d., hollow bars for cells that are not genuine, and OLMo-7B omitted as not re-extracted.

(a) The census of record on the 1000 ImageNet class prompts: padding-free extraction, Haar null, 99.9th-percentile statistic, 200 replicates.

| model          | type      | $d$  | $\hat{\delta}_{99.9}$                    | excess  | record<br>exc./null | $r/200$ | $p$  | $2 \times 2$ | cosine<br>exc. | $r$ |
|----------------|-----------|------|------------------------------------------|---------|---------------------|---------|------|--------------|----------------|-----|
| GPT-2 S        | causal LM | 768  | 0.077                                    | -0.013  | -15%                | 197     | .020 | ●○○○         | +0.000°        | 101 |
| GPT-2 M        | causal LM | 1024 | 0.052                                    | -0.005° | -9%                 | 170     | .154 | ○○○○         | -0.019         | 200 |
| GPT-2 L        | causal LM | 1280 | 0.053                                    | -0.023  | -30%                | 200     | .005 | ●●●●         | -0.016         | 200 |
| GPT-2 XL       | causal LM | 1600 | 0.048                                    | -0.027  | -36%                | 200     | .005 | ●●●●         | -0.018         | 200 |
| Pythia-410M    | causal LM | 1024 | 0.067                                    | -0.018  | -21%                | 200     | .005 | ●●●●         | -0.024         | 200 |
| Pythia-1B      | causal LM | 2048 | 0.066                                    | -0.015  | -18%                | 199     | .010 | ●●●●         | -0.026         | 200 |
| Pythia-2.8B    | causal LM | 2560 | 0.066                                    | -0.017  | -20%                | 200     | .005 | ●●●●         | -0.023         | 200 |
| OLMo-1B        | causal LM | 2048 | 0.051                                    | -0.024  | -32%                | 200     | .005 | ●●●●         | -0.032         | 200 |
| OLMo-7B        | causal LM | 4096 | not re-extracted (exceeds the local GPU) |         |                     |         |      |              |                |     |
| BGE-base       | embedder  | 768  | 0.078                                    | +0.003° | +4%                 | 26      | .871 | ○○○○         | -0.002°        | 151 |
| BGE-large      | embedder  | 1024 | 0.078                                    | +0.002° | +3%                 | 49      | .756 | ○○○○         | -0.003°        | 170 |
| GTE-base       | embedder  | 768  | 0.076                                    | +0.004° | +6%                 | 11      | .945 | ○○○○         | +0.001°        | 79  |
| GTE-large      | embedder  | 1024 | 0.075                                    | +0.003° | +3%                 | 36      | .821 | ○○○○         | -0.001°        | 131 |
| GTE-Qwen2-1.5B | embedder  | 1536 | 0.078                                    | +0.002° | +3%                 | 58      | .711 | ○○○○         | -0.006         | 199 |
| E5-base        | embedder  | 768  | 0.074                                    | +0.004° | +6%                 | 15      | .925 | ○○○○         | +0.000°        | 97  |
| E5-large       | embedder  | 1024 | 0.071                                    | +0.002° | +3%                 | 44      | .781 | ○○○○         | -0.001°        | 119 |

(b) Prompt templates: raw  $\hat{\delta}$  per template and, for the four GPT-2 sizes, the template-conditioned excess over the spectrum null (3 replicates; \*:  $|z| \geq 2$ ) and its fraction of the null.

| template                  | raw $\hat{\delta}$ |       |         |         | GPT-2 excess (padding-free) |         |         |          | GPT-2 excess / null |      |      |      |
|---------------------------|--------------------|-------|---------|---------|-----------------------------|---------|---------|----------|---------------------|------|------|------|
|                           | BGE-b              | E5-b  | GPT-2 S | GPT-2 M | S                           | M       | L       | XL       | S                   | M    | L    | XL   |
| photo                     | 0.124              | 0.114 | 0.131   | 0.091   | -0.016*                     | -0.019* | -0.043* | -0.050*  | -11%                | -19% | -34% | -41% |
| name_only                 | 0.118              | 0.119 | 0.145   | 0.122   | -0.006                      | -0.007  | -0.048* | -0.051*  | -6%                 | -5%  | -39% | -43% |
| image                     | 0.125              | 0.117 | 0.110   | 0.110   | -0.018*                     | -0.028* | -0.050* | -0.049*  | -13%                | -26% | -41% | -41% |
| closeup                   | 0.126              | 0.120 | 0.086   | 0.100   | +0.007                      | -0.018* | -0.044* | -0.049*  | +5%                 | -18% | -36% | -40% |
| this_is                   | 0.125              | 0.115 | 0.080   | 0.134   | -0.032*                     | -0.022* | -0.042* | -0.046*  | -19%                | -22% | -34% | -38% |
| wild                      | 0.115              | 0.116 | 0.089   | 0.104   | -0.023*                     | -0.067* | -0.014  | -0.010*  | -16%                | -43% | -11% | -8%  |
| def_pair                  | 0.119              | 0.108 | 0.105   | 0.081   | -0.002                      | -0.019* | -0.036* | -0.035*  | -1%                 | -21% | -28% | -29% |
| gloss_only                | 0.115              | 0.115 | 0.124   | 0.107   | -0.020*                     | -0.030* | -0.033* | -0.036*  | -14%                | -31% | -27% | -29% |
| concept                   | 0.119              | 0.120 | 0.112   | 0.112   | -0.002                      | -0.003  | -0.043* | -0.041*  | -1%                 | -3%  | -38% | -38% |
| discussion                | 0.121              | 0.116 | 0.105   | 0.098   | -0.010                      | -0.013* | -0.041* | -0.040*  | -7%                 | -17% | -35% | -35% |
| mean over templates       |                    |       |         |         | -0.012                      | -0.023  | -0.040  | -0.041   |                     |      |      |      |
| templates at $ z  \geq 2$ |                    |       |         |         | 5 of 10                     | 8 of 10 | 9 of 10 | 10 of 10 |                     |      |      |      |

(c) Extraction: the same 1000 prompts, only batch size and precision vary.

| batch size     | GPT-2 M             |                     | OLMo-1B             |                     |
|----------------|---------------------|---------------------|---------------------|---------------------|
|                | fp32 $\hat{\delta}$ | fp16 $\hat{\delta}$ | fp32 $\hat{\delta}$ | fp16 $\hat{\delta}$ |
| 1 (no padding) | 0.0816              | 0.0817              | 0.0817              | 0.0817              |
| 16             | 0.0996              | 0.0997              | 0.0817              | 0.0816              |
| 32             | 0.0905              | 0.0905              | 0.0817              | 0.0817              |

Table 11: **In text, clustered structure depends on recipe and scale: genuine at every Pythia scale and at the larger GPT-2 sizes, at the margin for GPT-2 S, absent for GPT-2 M and for the sentence embedders on this probe.** (a)  $r$  = replicates above the real value,  $p$  the left-tail add-one  $p$ -value; genuine = BH-corrected  $p \leq 0.05$  over the 15 models (°: not genuine); exc./null: excess as a fraction of the null reading; the  $2 \times 2$  column gives the verdicts under Haar $\times$ p99.9, Haar $\times$ supremum, Gaussian $\times$ p99.9, Gaussian $\times$ supremum (● genuine); cosine: the same census on cosine geometry. Genuine 7/15; GPT-2 S is genuine at  $p = .020$  with excess  $-0.013$  and GPT-2 M is not ( $p = .154$ ,  $-0.005$ ), while L and XL are genuine under every construction. GTE-base has the highest effective rank and is not genuine, so genuineness is not a proxy for anisotropy; the only LLM-backed embedder (GTE-Qwen2-1.5B) is not genuine. (b) Causal-LM values are template-sensitive (range of raw  $\hat{\delta}$  up to 0.065 for causal LMs against  $\leq 0.012$  for embedders; “gloss only” contains no class name) but the scale ordering is template-robust: with only 3 null replicates per cell the starred S and M cells are borderline, as the 200-replicate record in (a) shows, whereas the L/XL cells are not in doubt. (c) Precision is irrelevant (fp16 equals fp32 to four decimals); for GPT-2, left-padded batching changes the hidden states (mean cosine to the padding-free extraction 0.98) and moves  $\hat{\delta}$  by up to 0.018, while OLMo handles left padding correctly and is invariant, which is why padding-free extraction is the record.

## A.11 IS LOCAL STRUCTURE SHARED? PERMUTATION-CALIBRATED AGREEMENT

| measure                          | raw   | null mean | $\tau_{0.05}$ | calibrated | null-centered | pairs $p < 0.05$ |
|----------------------------------|-------|-----------|---------------|------------|---------------|------------------|
| mutual-kNN ( $k=10$ ), Euclidean | 0.472 | 0.010     | 0.015         | 0.464      | 0.462         | 66/66            |
| mutual-kNN, Poincaré             | 0.425 | 0.010     | 0.019         | 0.414      | 0.415         | 66/66            |
| linear CKA, Euclidean            | 0.633 | 0.108     | 0.111         | 0.577      | 0.524         | 66/66            |
| linear CKA, Poincaré             | 0.640 | 0.108     | 0.111         | 0.586      | 0.532         | 66/66            |

Table 12: **Cross-model agreement survives permutation calibration.** Means over the 66 model pairs of the 12 vision backbones on the 1000 ImageNet class centroids (census cache). Calibration of Gröger et al. (2026) with  $K=200$  permutations of the class correspondence and  $\alpha=0.05$ , scalar (no layer search):  $\tau_{0.05}$  is the  $\lceil 0.95(K+1) \rceil$ -th order statistic of the observed score and its nulls (eq. 9),  $p = (1 + \#\{\text{null} \geq \text{obs}\})/(K+1)$  (eq. 10), calibrated =  $\max\{(\text{obs} - \tau_{0.05})/(1 - \tau_{0.05}), 0\}$  (eq. 12); “null-centered” is the earlier raw-minus-null-mean variant, kept for continuity. The mKNN null mean equals the analytic chance level  $k/(n-1) = 10/999$ .

## A.12 WHAT FOLLOWS FOR PRACTICE? THE COROLLARY

**One parameter-free projection and two tasks serve the corollary.** The parameter-free map  $\phi(x) = \tanh(\frac{1}{2}\|s(x - \mu)\|) \cdot \frac{s(x - \mu)}{\|s(x - \mu)\|}$  centers features, scales the 95th-percentile norm to radius  $1/\sqrt{2}$ , and preserves directions; prototypes in the ball are means of projected points re-mapped through the origin. Tasks: nearest-centroid (NC) and 5-way 5-shot (FS, 1000 paired episodes); metric arms: Euclidean (R), Poincaré (H), the radial map with Euclidean distances (RT), cosine (COS), and the L2-normalized stack. Changing the readout metric on the intact configuration, which is what the rule selects among, is distinct from deforming the configuration itself.

**The calibrated reading predicts the gain within datasets.** Within each hierarchical dataset the Poincaré advantage tracks the record excess: the nearest-centroid correlation is  $-0.73$  on ImageNet with a confidence interval that excludes zero on all four datasets, the few-shot correlation excludes zero on CIFAR-100, CIFAR-10 and DTD but not on ImageNet, and the depth-test  $z$  predicts no gain anywhere (Table 13). The raw reading correlates as strongly ( $-0.84$  on ImageNet for nearest-centroid) and survives partialing out dimension, baseline accuracy, family and spectral covariates; without DINOv2 it holds on ImageNet and CIFAR-100 but not on CIFAR-10/DTD. Pooled cross-dataset correlations are family-driven and not used as evidence; on flat-label datasets no metric helps. Gains carry paired confidence intervals and a McNemar test.

**The training objective picks the metric.** The radial map with Euclidean distances changes nothing, so the gain is the metric itself, and the split follows the objective: cosine suffices for DINOv2, Poincaré  $\approx$  cosine for supervised ViTs, and Poincaré beats cosine for CLIP by up to  $+1.3\text{pp}$  even after L2 normalization (Figure 11; held-out selection and policy comparison in Table 13). The rule is read off the families it is scored on: it describes them rather than validating a policy.

**A diagnostic, not a policy.** Khrulkov et al. (2020) estimate  $c \approx 0.33$  from the raw reading of their datasets; the same rule on structureless Gaussian clouds gives the curvatures of Section 7. Cosine is a safe default, gating on the reading is the weakest policy tested, and gains are modest and limited to prototype-based tasks (sample-level tasks never benefit, Table 5). On DBpedia, a three-level text hierarchy, the diagnostic predicts correctly on both sides: genuine excess, no metric gain.

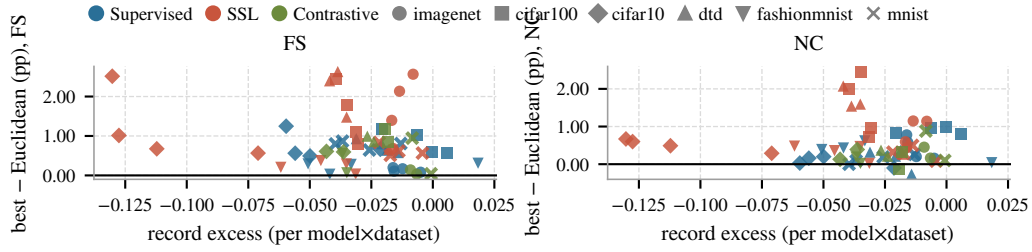


Figure 11: **Cells with a deeper calibrated excess carry larger zero-cost metric gains.** Plotted: best zero-cost metric minus Euclidean per cell against the record excess (FS left, NC right), one marker per dataset and family colors; the within-dataset correlations with confidence intervals are in Table 13.

(a) Task accuracies and zero-cost metric gains per cell (10 backbones with cached per-dataset features  $\times$  6 datasets).

| model    | data   | NC acc. (%) |        | FS acc. (%) |        | FS advantage over Euclidean (pp) |            |                           | McNemar        |
|----------|--------|-------------|--------|-------------|--------|----------------------------------|------------|---------------------------|----------------|
|          |        | Eucl.       | Poinc. | Eucl.       | Poinc. | cosine                           | radial map | Poinc. <sub>N</sub> - cos | NC H vs R      |
| ViT-T    | IN     | 68.1        | 67.2   | 97.5        | 97.6   | -0.01                            | +0.01      | +0.08                     | $10^{-22}$ (-) |
|          | C100   | 29.3        | 28.2   | 62.4        | 63.1   | +1.01                            | +0.45      | +0.55                     | $10^{-5}$ (-)  |
|          | C10    | 87.5        | 87.4   | 87.6        | 88.1   | +0.66                            | +0.19      | +0.29                     | 0.19 (-)       |
|          | DTD    | 63.7        | 62.5   | 83.3        | 83.9   | -0.23                            | +0.32      | +0.69                     | 0.03 (-)       |
|          | FMNIST | 78.5        | 78.4   | 81.6        | 81.9   | +0.21                            | +0.10      | +0.31                     | 0.38 (-)       |
|          | MNIST  | 75.9        | 74.0   | 74.3        | 74.6   | +0.64                            | +0.52      | +0.20                     | $10^{-13}$ (-) |
| ViT-S    | IN     | 77.7        | 77.5   | 98.6        | 98.7   | +0.17                            | -0.02      | -0.07                     | 0.09 (-)       |
|          | C100   | 74.9        | 74.6   | 92.2        | 92.4   | +0.59                            | +0.16      | -0.00                     | 0.10 (-)       |
|          | C10    | 93.4        | 93.2   | 92.0        | 92.5   | +0.56                            | +0.27      | +0.19                     | 0.02 (-)       |
|          | DTD    | 69.3        | 68.5   | 85.4        | 86.0   | -0.03                            | +0.25      | +0.73                     | 0.10 (-)       |
|          | FMNIST | 80.7        | 80.4   | 83.7        | 83.5   | +0.28                            | +0.08      | -0.19                     | 0.08 (-)       |
|          | MNIST  | 84.8        | 84.1   | 80.5        | 81.4   | +0.34                            | +0.65      | +0.84                     | $10^{-4}$ (-)  |
| ViT-B    | IN     | 81.6        | 81.7   | 98.9        | 99.0   | +0.20                            | -0.04      | -0.05                     | 0.21 (+)       |
|          | C100   | 77.3        | 77.0   | 92.5        | 92.7   | +0.56                            | +0.12      | -0.05                     | 0.07 (-)       |
|          | C10    | 94.5        | 94.0   | 92.7        | 93.0   | +0.49                            | +0.18      | +0.15                     | $10^{-5}$ (-)  |
|          | DTD    | 70.8        | 70.8   | 85.4        | 86.2   | +0.28                            | +0.19      | +0.69                     | 1.00 (-)       |
|          | FMNIST | 81.3        | 81.6   | 83.4        | 83.3   | +0.03                            | +0.09      | -0.04                     | 0.08 (+)       |
|          | MNIST  | 81.6        | 80.5   | 76.8        | 77.6   | +0.19                            | +0.51      | +0.66                     | $10^{-7}$ (-)  |
| ViT-L    | IN     | 86.1        | 86.2   | 99.4        | 99.4   | +0.11                            | -0.03      | -0.02                     | 0.58 (+)       |
|          | C100   | 85.3        | 85.2   | 94.8        | 95.5   | +1.17                            | +0.31      | +0.12                     | 0.59 (-)       |
|          | C10    | 97.5        | 97.5   | 95.0        | 95.8   | +1.25                            | +0.35      | +0.59                     | 0.76 (+)       |
|          | DTD    | 71.8        | 71.5   | 85.6        | 86.5   | +0.13                            | +0.16      | +1.02                     | 0.69 (-)       |
|          | FMNIST | 83.7        | 83.5   | 84.4        | 84.7   | +0.00                            | +0.17      | +0.23                     | 0.12 (-)       |
|          | MNIST  | 84.2        | 83.6   | 80.8        | 81.6   | +0.08                            | +0.42      | +0.67                     | 0.00 (-)       |
| DINOv2-S | IN     | 74.7        | 74.4   | 97.4        | 97.9   | +0.68                            | +0.03      | -0.22                     | 0.00 (-)       |
|          | C100   | 73.4        | 74.4   | 94.3        | 95.1   | +0.80                            | +0.15      | +0.06                     | $10^{-6}$ (+)  |
|          | C10    | 95.0        | 95.2   | 92.9        | 93.5   | -0.36                            | +0.13      | +1.06                     | 0.14 (+)       |
|          | DTD    | 73.4        | 73.4   | 89.8        | 90.5   | +0.30                            | +0.04      | +0.36                     | 1.00 (+)       |
|          | FMNIST | 82.2        | 82.2   | 85.4        | 85.5   | +0.02                            | +0.07      | +0.07                     | 1.00 (-)       |
|          | MNIST  | 83.4        | 81.9   | 80.2        | 80.8   | -0.07                            | +0.45      | +0.67                     | $10^{-11}$ (-) |
| DINOv2-B | IN     | 78.9        | 78.9   | 97.0        | 97.8   | +1.40                            | +0.04      | -0.63                     | 0.57 (+)       |
|          | C100   | 87.4        | 87.9   | 96.5        | 97.2   | +1.11                            | +0.09      | -0.34                     | $10^{-4}$ (+)  |
|          | C10    | 97.1        | 97.4   | 94.7        | 95.4   | +0.27                            | +0.03      | +0.43                     | $10^{-4}$ (+)  |
|          | DTD    | 74.2        | 74.7   | 89.2        | 90.3   | +1.48                            | +0.01      | -0.35                     | 0.31 (+)       |
|          | FMNIST | 83.2        | 82.8   | 85.9        | 85.9   | +0.21                            | +0.04      | -0.18                     | 0.01 (-)       |
|          | MNIST  | 80.6        | 78.8   | 76.6        | 77.1   | +0.17                            | +0.48      | +0.47                     | $10^{-16}$ (-) |
| DINOv2-L | IN     | 79.8        | 80.4   | 96.2        | 97.1   | +2.13                            | +0.02      | -1.24                     | $10^{-14}$ (+) |
|          | C100   | 85.6        | 87.7   | 96.1        | 97.1   | +1.80                            | +0.09      | -0.74                     | $10^{-38}$ (+) |
|          | C10    | 97.9        | 98.3   | 94.1        | 95.1   | +0.92                            | -0.00      | +0.11                     | $10^{-8}$ (+)  |
|          | DTD    | 74.4        | 75.2   | 87.1        | 88.4   | +2.40                            | +0.03      | -0.98                     | 0.11 (+)       |
|          | FMNIST | 84.4        | 84.5   | 87.8        | 87.9   | +0.29                            | +0.03      | -0.03                     | 0.40 (+)       |
|          | MNIST  | 74.1        | 72.7   | 70.5        | 71.1   | +0.25                            | +0.43      | +0.46                     | $10^{-12}$ (-) |
| DINOv2-G | IN     | 78.4        | 78.3   | 96.2        | 97.1   | +2.57                            | +0.03      | -1.65                     | 0.10 (-)       |
|          | C100   | 89.8        | 91.0   | 95.9        | 97.2   | +2.44                            | +0.13      | -1.10                     | $10^{-16}$ (+) |
|          | C10    | 97.8        | 98.2   | 92.3        | 93.8   | +2.51                            | -0.02      | -1.01                     | $10^{-13}$ (+) |
|          | DTD    | 72.9        | 73.1   | 87.2        | 88.5   | +2.63                            | +0.01      | -1.38                     | 0.81 (+)       |
|          | FMNIST | 85.9        | 85.8   | 89.0        | 89.2   | +0.38                            | +0.04      | -0.07                     | 0.72 (-)       |
|          | MNIST  | 79.3        | 78.6   | 72.9        | 73.8   | +0.30                            | +0.56      | +0.72                     | 0.00 (-)       |
| CLIP-B   | IN     | 61.4        | 59.9   | 96.8        | 96.8   | -0.04                            | -0.00      | +0.14                     | $10^{-40}$ (-) |
|          | C100   | 65.6        | 65.1   | 87.7        | 88.9   | -0.03                            | +0.44      | +1.21                     | 0.07 (-)       |
|          | C10    | 87.9        | 88.3   | 89.5        | 90.1   | -0.19                            | +0.20      | +0.90                     | $10^{-4}$ (+)  |
|          | DTD    | 63.7        | 63.2   | 82.2        | 83.1   | -0.23                            | +0.36      | +1.29                     | 0.46 (-)       |
|          | FMNIST | 75.9        | 75.1   | 78.8        | 78.9   | +0.06                            | +0.17      | +0.10                     | $10^{-4}$ (-)  |
|          | MNIST  | 70.1        | 66.6   | 70.9        | 69.7   | +0.04                            | +0.23      | -1.17                     | $10^{-45}$ (-) |
| CLIP-L   | IN     | 75.0        | 74.5   | 98.4        | 98.6   | +0.01                            | +0.00      | +0.22                     | $10^{-8}$ (-)  |
|          | C100   | 74.6        | 74.5   | 90.0        | 90.8   | +0.12                            | +0.39      | +1.23                     | 0.66 (-)       |
|          | C10    | 95.9        | 95.9   | 92.4        | 93.0   | +0.16                            | +0.25      | +0.91                     | 0.59 (-)       |
|          | DTD    | 68.6        | 69.0   | 85.2        | 86.2   | -0.08                            | +0.36      | +1.24                     | 0.45 (+)       |
|          | FMNIST | 76.2        | 75.9   | 76.8        | 77.3   | +0.12                            | +0.33      | +0.68                     | 0.09 (-)       |
|          | MNIST  | 85.2        | 86.1   | 83.7        | 84.7   | +0.08                            | +0.58      | +1.06                     | $10^{-7}$ (+)  |

Table 13: **The calibrated reading predicts the zero-cost gain within datasets, the depth verdict predicts nothing, and cosine is the safe default.** (a) Nearest-centroid (NC) and 5-way 5-shot (FS, 1000 paired episodes) accuracy under Euclidean and Poincaré readouts, regenerated from the audited reruns; FS advantage of cosine and of the radial map with Euclidean distances (the map without the metric changes nothing) over Euclidean, and of the Poincaré readout over cosine after L2 normalization; McNemar  $p$  for the NC test-set decisions, Poincaré vs Euclidean (order of magnitude when  $p < 10^{-3}$ ; sign of the NC advantage in parentheses).

(b) Correlation between the raw supremum  $\delta_{\text{norm}}$  and the metric gain within each dataset ( $n=10$  backbones), with bootstrap CI95 and the family-identity control (demeaned: both variables centered within family; within-family  $r$  for ViT and DINOv2, sign only for CLIP).

| gain      | dataset                    | $r$ [CI95]           | demeaned | ViT   | DINOv2 | CLIP |
|-----------|----------------------------|----------------------|----------|-------|--------|------|
| NC H—R    | ImageNet                   | −0.83 [−0.97, −0.65] | −0.73    | −0.99 | −0.77  | (−1) |
|           | CIFAR-100                  | −0.87 [−0.97, −0.78] | −0.45    | −0.68 | −0.33  | (−1) |
|           | CIFAR-10                   | −0.63 [−0.97, −0.06] | −0.31    | +0.01 | −0.98  | (+1) |
|           | DTD                        | −0.46 [−0.90, +0.22] | −0.57    | −0.78 | −0.76  | (+1) |
|           | pooled (cluster bootstrap) | −0.45 [−0.60, −0.36] | —        | —     | —      | —    |
| FS H—R    | ImageNet                   | −0.57 [−0.91, +0.14] | −0.73    | −0.04 | −0.97  | (−1) |
|           | CIFAR-100                  | −0.67 [−0.95, −0.15] | −0.63    | −0.41 | −0.96  | (+1) |
|           | CIFAR-10                   | −0.78 [−0.93, −0.23] | −0.64    | −0.22 | −0.79  | (−1) |
|           | DTD                        | −0.71 [−0.94, −0.32] | −0.86    | −0.95 | −0.98  | (+1) |
|           | pooled (cluster bootstrap) | −0.18 [−0.50, +0.12] | —        | —     | —      | —    |
| FS best—R | ImageNet                   | −0.64 —              | −0.71    | −0.57 | −0.90  | (−1) |
|           | CIFAR-100                  | −0.83 —              | −0.82    | −0.58 | −0.96  | (+1) |
|           | CIFAR-10                   | −0.63 —              | −0.55    | −0.26 | −0.65  | (−1) |
|           | DTD                        | −0.60 —              | −0.70    | −0.95 | −0.88  | (+1) |

(c) Best zero-cost metric per backbone: few-shot advantage over Euclidean, mean over the four hierarchical datasets, and which metric collects it (either: within 0.15pp).

| model    | best—R (pp) | metric |
|----------|-------------|--------|
| ViT-T    | +0.58       | either |
| ViT-S    | +0.49       | either |
| ViT-B    | +0.51       | either |
| ViT-L    | +0.87       | either |
| DINOv2-S | +0.68       | H      |
| DINOv2-B | +1.17       | cos    |
| DINOv2-L | +1.84       | cos    |
| DINOv2-G | +2.54       | cos    |
| CLIP-B   | +0.67       | H      |
| CLIP-L   | +0.65       | H      |

Table 13: (continued) (b) Within-dataset correlations of the raw supremum reading; bootstrap CI95 from 10k resamples of the 10 models, pooled rows by cluster bootstrap over datasets; family control: both variables centered within family before correlating, and within-family correlations (ViT, DINOv2: 4 models each; CLIP has two, so only its sign is shown). Without DINOv2 (six backbones) the raw NC correlation holds on ImageNet and CIFAR-100 (−0.87/−0.65) but not on CIFAR-10/DTD (+0.23/−0.34); pooled cross-dataset correlations of the best-metric advantage do not survive the family control (−0.31 → −0.18 over all 60 cells, −0.33 → −0.04 on hierarchical cells) and are family-driven. On the record excess the nearest-centroid prediction survives (CI95 excluding zero) on ImageNet, CIFAR-100, CIFAR-10, DTD and the few-shot prediction on CIFAR-100, CIFAR-10, DTD; the depth verdict predicts no gain in any dataset. (c) The gain per backbone and which readout collects it.

(d) The same correlations on the raw  $\hat{\delta}_{99.9}$ , the record excess and the depth-test  $z$  (Fisher- $z$  CI95; dm: family-demeaned  $r$ ).

| gain      | dataset   | raw $\hat{\delta}_{99.9}$ |       | record excess        |       | depth $z$            |  |
|-----------|-----------|---------------------------|-------|----------------------|-------|----------------------|--|
|           |           | $r$ [CI95]                | dm    | $r$ [CI95]           | dm    | $r$ [CI95]           |  |
| NC H—R    | ImageNet  | −0.84 [−0.96, −0.45]      | −0.83 | −0.73 [−0.93, −0.19] | −0.61 | −0.28 [−0.77, +0.42] |  |
|           | CIFAR-100 | −0.92 [−0.98, −0.68]      | −0.58 | −0.79 [−0.95, −0.33] | −0.21 | +0.52 [−0.16, +0.87] |  |
|           | CIFAR-10  | −0.63 [−0.90, +0.00]      | −0.31 | −0.67 [−0.91, −0.06] | −0.34 | n/a                  |  |
|           | DTD       | −0.67 [−0.91, −0.07]      | −0.67 | −0.76 [−0.94, −0.26] | −0.60 | n/a                  |  |
| FS H—R    | ImageNet  | −0.86 [−0.97, −0.50]      | −0.56 | −0.21 [−0.74, +0.48] | +0.35 | +0.27 [−0.43, +0.77] |  |
|           | CIFAR-100 | −0.62 [−0.90, +0.02]      | −0.51 | −0.79 [−0.95, −0.31] | −0.80 | −0.26 [−0.76, +0.45] |  |
|           | CIFAR-10  | −0.78 [−0.95, −0.31]      | −0.64 | −0.78 [−0.95, −0.30] | −0.66 | n/a                  |  |
|           | DTD       | −0.87 [−0.97, −0.53]      | −0.94 | −0.98 [−1.00, −0.91] | −0.97 | n/a                  |  |
| FS best—R | ImageNet  | −0.83 [−0.96, −0.43]      | −0.50 | −0.08 [−0.68, +0.58] | +0.47 | +0.13 [−0.54, +0.70] |  |
|           | CIFAR-100 | −0.77 [−0.94, −0.27]      | −0.75 | −0.75 [−0.94, −0.23] | −0.65 | +0.22 [−0.48, +0.75] |  |
|           | CIFAR-10  | −0.63 [−0.90, −0.01]      | −0.55 | −0.62 [−0.90, +0.02] | −0.56 | n/a                  |  |
|           | DTD       | −0.81 [−0.95, −0.36]      | −0.78 | −0.90 [−0.98, −0.63] | −0.84 | n/a                  |  |

(e) Few-shot metric-selection policies on held-out episodes (500 validation / 500 test per cell).

| policy (test advantage over Euclidean, pp) | all 60 cells | hierarchical (40) |
|--------------------------------------------|--------------|-------------------|
| always cosine                              | +0.23        | +0.28             |
| objective rule (zero validation)           | +0.28        | +0.41             |
| $\delta$ -gated objective rule             | +0.08        | +0.13             |
| validation-picked per cell                 | +0.63        | +0.78             |
| test oracle (reference)                    | +0.64        | +0.78             |

(f) Nearest-centroid metric-selection policies.

| NC policy (advantage over Euclidean, pp) | all 60 | flat (20) | hierarchical (40) |
|------------------------------------------|--------|-----------|-------------------|
| always cosine                            | +0.48  | +0.24     | +0.59             |
| always Poincaré                          | −0.24  | −0.70     | −0.00             |
| gated rule (flat→R)                      | +0.34  | +0.00     | +0.51             |
| $\delta$ -gated rule                     | +0.19  | +0.00     | +0.29             |

(g) Sensitivity to the projection target  $t$  (36 hierarchical cells).

| target $t$        | 0.25  | 0.35  | 0.41  | 0.45  | 0.50  | 0.58  | 0.71  | 0.95  |
|-------------------|-------|-------|-------|-------|-------|-------|-------|-------|
| FS adv. mean (pp) | +0.88 | +0.87 | +0.87 | +0.86 | +0.84 | +0.82 | +0.76 | +0.56 |
| NC adv. mean (pp) | −0.30 | −0.25 | −0.23 | −0.21 | −0.17 | −0.12 | −0.01 | +0.15 |
| % cells FS > 0    | 97    | 100   | 100   | 100   | 100   | 100   | 100   | 100   |

Table 13: (continued) (d) The calibrated reading: the raw 99.9th-percentile statistic, the record excess and the depth-test  $z$  of Table 6 as predictors. (e) Validation picking retains the oracle advantage and the zero-validation objective rule modestly beats always-cosine. (f) The gate’s value is avoiding the Poincaré projection on flat-label data; cosine is a safe NC default everywhere. (g) The headline  $t = 1/\sqrt{2} \approx 0.71$  was fixed a priori and is conservative for FS.

### A.13 DOES $\xi$ READ CURVATURE? THE PARALLELOGRAM DEFECT AS A NORM-STRUCTURE DESCRIPTOR

**$\xi$  reads norm structure, not curvature.** A parallelogram estimator  $\xi$ , intended as a curvature sign, is driven negative by heterogeneous norms alone, so it is reported only as a norm-structure descriptor. For a triangle  $(a, b, c)$  with  $m$  the approximate midpoint of  $(b, c)$  (the minimizer of  $D(\cdot, b) + D(\cdot, c) + |D(\cdot, b) - D(\cdot, c)|$ ), the parallelogram defect is  $\xi = (D(a, m)^2 + \frac{1}{4}D(b, c)^2 - \frac{1}{2}D(a, b)^2 - \frac{1}{2}D(a, c)^2) / (2 D(a, m) \text{diam})$ , averaged over  $2 \times 10^4$  sampled triangles: 0 in Euclidean space, positive on spheres, negative in hyperbolic space and on trees.

(a)  $\xi$  on reference geometries and ImageNet centroids, with the spectrum-matched null (20 replicates).

| space / model                   | $\xi$ (mean) | frac. negative | null $\xi$         | excess ( $\pm$ CI95) | $z$ |
|---------------------------------|--------------|----------------|--------------------|----------------------|-----|
| iid Gaussian ( $d=768$ )        | +0.092       | 0%             | —                  | —                    | —   |
| $S^d$ (geodesic)                | +0.113       | 0%             | —                  | —                    | —   |
| $S^d$ (chord)                   | +0.115       | 0%             | —                  | —                    | —   |
| $\mathbb{H}^2$ region ( $R=6$ ) | -0.194       | 96%            | —                  | —                    | —   |
| balanced tree (depth 9)         | -0.239       | 96%            | —                  | —                    | —   |
| ViT-T                           | +0.027       | 25%            | +0.027 $\pm$ 0.002 | -0.001 $\pm$ 0.003   | -0  |
| ViT-S                           | +0.027       | 23%            | +0.042 $\pm$ 0.002 | -0.015 $\pm$ 0.005   | -7  |
| ViT-B                           | +0.033       | 16%            | +0.047 $\pm$ 0.002 | -0.014 $\pm$ 0.004   | -6  |
| ViT-L                           | +0.028       | 21%            | +0.046 $\pm$ 0.002 | -0.018 $\pm$ 0.005   | -7  |
| DINO-B                          | -0.008       | 55%            | +0.052 $\pm$ 0.003 | -0.059 $\pm$ 0.005   | -21 |
| DINOv2-S                        | -0.020       | 66%            | +0.061 $\pm$ 0.004 | -0.081 $\pm$ 0.008   | -20 |
| DINOv2-B                        | -0.049       | 83%            | +0.075 $\pm$ 0.003 | -0.123 $\pm$ 0.005   | -45 |
| DINOv2-L                        | -0.068       | 90%            | +0.080 $\pm$ 0.002 | -0.149 $\pm$ 0.004   | -66 |
| DINOv2-G                        | -0.070       | 89%            | +0.082 $\pm$ 0.002 | -0.152 $\pm$ 0.004   | -69 |
| CLIP-B                          | +0.019       | 32%            | +0.024 $\pm$ 0.002 | -0.005 $\pm$ 0.003   | -3  |
| CLIP-L                          | +0.028       | 28%            | +0.032 $\pm$ 0.002 | -0.004 $\pm$ 0.004   | -2  |
| SigLIP-B                        | +0.036       | 20%            | +0.031 $\pm$ 0.002 | +0.005 $\pm$ 0.004   | +2  |

(b)  $\xi$  on the angular geometry: centroids L2-normalized, spherical geodesic distances, null on the normalized cloud.

| model    | $\xi_{\text{geo}}$ | frac. neg. | excess  | $z$   |
|----------|--------------------|------------|---------|-------|
| ViT-T    | +0.031             | 0.24       | -0.0048 | -2.7  |
| ViT-S    | +0.065             | 0.08       | +0.0092 | +6.2  |
| ViT-B    | +0.067             | 0.07       | +0.0144 | +10.5 |
| ViT-L    | +0.074             | 0.04       | +0.0113 | +7.9  |
| DINO-B   | +0.085             | 0.05       | +0.0014 | +1.0  |
| DINOv2-S | +0.100             | 0.02       | +0.0033 | +2.3  |
| DINOv2-B | +0.108             | 0.01       | +0.0046 | +4.9  |
| DINOv2-L | +0.108             | 0.01       | +0.0040 | +3.9  |
| DINOv2-G | +0.094             | 0.01       | -0.0114 | -12.5 |
| CLIP-B   | +0.025             | 0.28       | -0.0129 | -7.1  |
| CLIP-L   | +0.033             | 0.25       | -0.0120 | -4.8  |
| SigLIP-B | +0.051             | 0.14       | +0.0025 | +1.2  |

(c)  $\xi$  on references with heterogeneous norms but no curvature ( $n=1000$ ,  $d=768$ , 5 seeds).

| reference (no curvature unless stated)     | $\xi$   | frac. negative |
|--------------------------------------------|---------|----------------|
| iid Gaussian                               | +0.0919 | 0.00           |
| Gaussian, lognormal radii ( $\sigma=0.3$ ) | -0.0621 | 0.99           |
| Gaussian, lognormal radii ( $\sigma=0.6$ ) | -0.0607 | 1.00           |
| Gaussian, lognormal radii ( $\sigma=1.0$ ) | -0.0496 | 0.99           |
| core + shell mixture                       | -0.3217 | 0.79           |
| uniform sphere                             | +0.1154 | 0.00           |
| sphere with radial jitter                  | -0.0571 | 1.00           |

Table 14:  $\xi$  reads norm structure, not curvature. (a) Matched nulls have positive  $\xi$  (noise combines null and estimator s.d.), so raw readings understate curvature. (b) The deep negative  $\xi$ -excess of DINOv2 on ImageNet is a norm-structure effect for S/B/L, whose angular  $\xi$ -excess turns sphere-leaning positive, while DINOv2-G and CLIP-B/L remain negative on the sphere. (c) Norm dispersion alone drives  $\xi$  negative (lognormal radii, radial jitter on a sphere, core+shell), so the estimator is reported only as a norm-structure descriptor.

## A.14 THE MODEL PANEL AND EXTRACTION

| type                    | model                              | params       |
|-------------------------|------------------------------------|--------------|
| Supervised ViT (vision) | ViT-T (i21k)                       | 5M           |
| Supervised ViT (vision) | ViT-S (i21k)                       | 22M          |
| Supervised ViT (vision) | ViT-B (i21k)                       | 86M          |
| Supervised ViT (vision) | ViT-L (i21k)                       | 307M         |
| SSL (vision)            | DINO-B                             | 86M          |
| SSL (vision)            | DINOv2-S                           | 22M          |
| SSL (vision)            | DINOv2-B                           | 86M          |
| SSL (vision)            | DINOv2-L                           | 307M         |
| SSL (vision)            | DINOv2-G                           | 1.1B         |
| Contrastive (vision)    | CLIP-B                             | 86M          |
| Contrastive (vision)    | CLIP-L                             | 307M         |
| Contrastive (vision)    | SigLIP-B                           | 86M          |
| Causal LM               | GPT-2 S                            | 117M         |
| Causal LM               | GPT-2 M                            | 345M         |
| Causal LM               | GPT-2 L                            | 774M         |
| Causal LM               | GPT-2 XL                           | 1.5B         |
| Causal LM               | Pythia-410M                        | 410M         |
| Causal LM               | Pythia-1B                          | 1.0B         |
| Causal LM               | Pythia-2.8B                        | 2.8B         |
| Causal LM               | OLMo-1B                            | 1.0B         |
| Causal LM               | OLMo-7B                            | 7.0B         |
| Text embedder           | BGE-base                           | 110M         |
| Text embedder           | BGE-large                          | 335M         |
| Text embedder           | GTE-base                           | 110M         |
| Text embedder           | GTE-large                          | 335M         |
| Text embedder           | GTE-Qwen2-1.5B                     | 1.5B         |
| Text embedder           | E5-base                            | 110M         |
| Text embedder           | E5-large                           | 335M         |
| Controls (vision)       | DeiT-B/16 (IN-1k, no distillation) | 86M          |
| Controls (vision)       | ViT-B/16 augreg (IN-1k)            | 86M          |
| Controls (vision)       | MERU ViT-S/B/L and CLIP twins      | 22M/86M/307M |
| Controls (vision)       | Barlow Twins, BYOL (ResNet-50)     | 24M          |

Table 15: **The model panel.** 12 vision backbones (all in the census; the 10 with cached per-dataset features, all but DINO-B and SigLIP-B, form the task grid), 9 causal LMs, 7 text embedders, and the vision models used only as controls. Class centroids average 100 cached training images per class on ImageNet and the full cached split, 80–6,000 images per class, on the other datasets. Feature extraction: vision backbones use the encoder’s default pooled embedding (timm, num\_classes=0, CLS or global pool per architecture); causal LMs use the last-token hidden state, extracted one prompt at a time (padding-free: left-padded batching alters GPT-2’s hidden states, Table 11); text embedders use their recommended pooling (CLS for BGE, mean for GTE/E5, last token for GTE-Qwen2); CLIP/SigLIP text towers use their text encoders.

## A.15 PROVENANCE

| table | generated file        | result file(s)                                                                                                                                                                                                                                                                                                                                   |
|-------|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3     | tab_q01_census        | expR52_census_haar_p999_200.csv<br>expR54_census_haar_sup_200.csv expR40b_p999census200.csv<br>expR39c_census200_cache.csv<br>expR57_census_cosine_haar_p999_200.csv<br>expR57_text_cosine_haar_p999_200.csv<br>exp10_local_vs_global.csv expR45_convnet_rows.csv<br>exp20_null_ztable.csv                                                       |
| 11    | tab_q02_text          | expR53_text_haar_p999_200.csv expR53_text_haar_sup_200.csv<br>expR53_text_gauss_p999_200.csv expR48b_text_census200_bs1.csv<br>expR57_text_cosine_haar_p999_200.csv exp18_text_nulls.csv<br>exp4_prompt_variation.csv exp17_wordnet_prompts.csv<br>expR49_template_nulls_bs1.csv expR47_extraction_variance.csv<br>expR62_samplelevel_record.csv |
| 5     | tab_q03_sample        | expR56_depth_variants.csv expR69_depth_haarhubs_summary.csv                                                                                                                                                                                                                                                                                      |
| 6     | tab_q04_depth         | expR64b_wn30bal_summary.csv expR70_inet1k_supervised.csv<br>expR52_census_haar_p999_200.csv expR63_meru_record.csv<br>expR65_hier_finetune.csv expR71_meru_radii.csv                                                                                                                                                                             |
| 7     | tab_q05_power         | expR55_depth_power.csv expR55b_depth_power_leafframe.csv<br>expR64b_wn30.csv expR64b_wn30_summary.csv<br>expR64b_wn30bal_summary.csv expR67_frame_choice.json<br>expR69_depth_haarhubs_summary.csv                                                                                                                                               |
| 9     | tab_q06_treemap       | exp22_tree_similarity_imagenet.npz exp23_treemap_controls.npz<br>exp23_config_diagnostics.json<br>expR58_treemap_cutfree_summary.csv                                                                                                                                                                                                             |
| 10    | tab_q07_wordnet       | exp3_alignment.csv exp28_recovery_per_config.csv<br>exp8_p1_recovery.csv exp1_delta_controls.csv<br>exp8_p6_pooling.csv expR70_inet1k_supervised.csv<br>exp14_dbpedia.csv expR61_dbpedia_record.csv<br>exp27_dbpedia_treemap.json exp16_hierarcaps.csv                                                                                           |
| 4     | tab_q08_robust        | expR72_budget_record.csv expR72_budget_record_summary.csv<br>expR31_quad_sweep.csv expR32_centroid_bootstrap.csv<br>expR46_null_variants.csv<br>expR59_imagenet_bootstrap_summary.csv<br>expR66_joint_sensitivity_summary.csv<br>expR60_c_sweep_record.csv                                                                                       |
| 13    | tab_q09_corollary     | table1_regenerated.csv exp2_metric_controls.csv<br>exp2b_normalized_stack.csv exp13_mcnemar.csv<br>night/correlation_cis.csv exp20_null_ztable.csv<br>expR68_corollary_excess.csv exp24_val_metric_selection.csv<br>night/t_sweep.csv                                                                                                            |
| 2     | tab_q10_calibration   | exp1_delta_controls.csv expR42_star_calibration.csv                                                                                                                                                                                                                                                                                              |
| 8     | tab_q11_interventions | analysis4_finetuning.csv e1_delta_by_layer.csv<br>e5_random_control.csv night/arch_matched_triplet.csv<br>night/orc_bridges.csv                                                                                                                                                                                                                  |
| 14    | tab_q12_xi            | exp12_curvature_sign.csv exp26_xi_nulls.csv<br>expR38_xi_angular.csv expR43_xi_norm_references.csv                                                                                                                                                                                                                                               |
| 12    | tab_q13_local         | exp21b_local_global_K200.csv                                                                                                                                                                                                                                                                                                                     |
| 15    | tab_q14_panel         | (model panel: stated in the generator; parameter counts<br>from the model cards)                                                                                                                                                                                                                                                                 |

Table 16: Provenance index: every appendix table is generated by script from the released per-cell result files listed here; no table value is hand-transcribed. The same mapping for main-text tables and figures is annotated in the source and shipped with the code release.